Module M1, Session by Session: Convex Sets, Separation and Farkas' Lemma

MMC007 — Optimization

MSc in Mathematical and Computational Modeling
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1 Purpose and place in the plan

M1 is the first module of the MMC007 reading plan and the only one with no prerequisites. It covers the first item of the syllabus — "properties of convex sets" — and it answers a question that item normally leaves unasked: *why does one geometric fact about convex sets generate every duality theorem in the course?*

The answer is that a point outside a closed convex set can be strictly separated from it by a hyperplane; that the separating vector is not an abstract existence but a vector a projection computes; and that every certificate of optimality or of infeasibility in the rest of the plan is an instance of that one construction. Three questions become askable at once, and this module answers all three:

1. Given a point and a convex set, how is the separating hyperplane *produced*? Answered by the projection theorem (S3) and its two-line corollary (S4).
2. When does separation *fail*? Answered by S2 and by the capstone of S7: it fails without closedness, and every duality gap in the course is a failure of exactly this kind.
3. What does separation give when the set is a finitely generated cone? Answered by Farkas' lemma (S6), which is the single result the rest of the plan quotes most often.

Every later module inherits something specific from M1:

- The *projection theorem* of S3 reappears in M8 as the stationarity condition $x^\star = P_C(x^\star - t\nabla f(x^\star))$ of the projected gradient method, and its firm nonexpansiveness is what makes the proximal iterates of M9 Fejér monotone. It is also, with C an affine subspace, the projection theorem MMC201's module M2 uses for least squares: one theorem, two courses, two guises.
- The *separation theorem* of S4 is the engine of M4. The proof of strong duality under Slater's condition separates the point $(0, p^\star)$ from the perturbation epigraph, and it is the same proof written here.
- *Farkas' lemma*, proved in S6, is repeated in M4 with the tangent cone in place of $\mathrm{cone}(A)$ — and the result is the Karush–Kuhn–Tucker theorem — and again in M11 with the constraint cone of a linear program, where the result is LP duality. *If those three proofs do not look like one proof when M11 is reached, M1 was not finished.*
- The *polar cone* and *Moreau's decomposition* of S5 are the normal cone M4 uses to write optimality conditions, and the conic special case of the proximal decomposition proved in M3.
- The *simplex projection* built in S3 is used verbatim by M8, M9 and M15; it is the operation every projected policy-gradient step performs.
- The *closedness lemma* of S6 is why the polyhedral cone of M11 is closed, hence why linear programming has no duality gap and needs no constraint qualification.

2 Learning objectives

On completing M1 the student should be able to do the following, without notes:

1. State and prove Carathéodory's theorem and its conic version, and say where finite dimension is used.
2. Distinguish the interior from the relative interior, exhibit a convex set with empty interior and nonempty relative interior, and say why every constraint qualification in the plan is stated with ri.
3. State and prove the projection theorem: existence, uniqueness, the variational characterization, nonexpansiveness and firm nonexpansiveness.
4. Derive the projection onto a halfspace, an affine subspace, a ball, a box and the probability simplex, and explain why clipping and renormalizing is *not* the projection onto the simplex.
5. State the three separation theorems — proper, strong, supporting — with their three different hypotheses, and prove strong separation from the projection theorem in two lines, exhibiting the separating vector explicitly.
6. Give two disjoint closed convex sets admitting no strong separation, and name the hypothesis that fails.
7. Define the polar cone, state and prove the bipolar theorem, and prove Moreau's decomposition, identifying the orthogonal decomposition of a subspace as its special case.
8. Prove that a finitely generated cone is closed, and exhibit a convex cone that is not.
9. State and prove Farkas' lemma in both its equality and inequality forms, marking the single line where closedness is used, and derive Gordan's theorem from it.
10. Say, in one sentence for each of convexity, closedness and finite generation, which theorem it enables and what breaks without it.

3 How to use this document

This document is self-contained. Every definition it uses is stated in it, every theorem it uses is proved in it, and no proof is left as "see the reference". There are exactly two exceptions, both marked in place: Proposition 7(2) and Theorem 18, which are stated without proof because they are needed only for orientation and their proofs are long and unrewarding at this stage. The books listed in Table 2 are for a *second view* after a session, not preparation before it. You should be able to work through M1 with this document, paper, and a Python interpreter.

The order within a session is deliberate. Following the expository method of Zhao's *Mathematical Foundations of Reinforcement Learning*, each session begins with the question it answers and an informal picture of the answer, states the result next, and only then proves it. Read in that order. A proof read before the statement it proves has been understood as a claim about a picture is a proof that will not be remembered.

Each of the seven sections below is one working session, and each has these parts, always in this order:

The question
: one paragraph. What is not yet known, stated so that the session's theorem is recognizably its answer.

The picture
: the answer in words and geometry, before any formalism. If you cannot draw this, do not proceed to the statements.

Statements and proofs
: definitions, theorems and complete proofs. These are to be read once, then reproduced on paper with the document closed.

Worked example
: one instance carried through with explicit numbers, so that the theorem becomes something you have computed rather than something you have read.

Where this is used later
: the forward links, named by module.

Proof of concept
: one implementation, sized to the session, with numbered validation checks. Every check is an inequality against a stated tolerance or an exact identity. A check reported without a number has not been performed.

Exercises written work, tagged where a later module reuses it.

Session gate
: the condition for starting the next session.

The gate rule. Do not start session $k+1$ until the gate of session k has been passed. The gates are cumulative by design: S7's capstone runs every earlier session's code again, and if an earlier check was skipped it will fail there instead.

Programming rules, once, for all sessions. Python with NumPy and SciPy. Every object in this module is one you construct, so every target is known in closed form or by an independent solve; nothing is estimated from data. Fix and report one seed for the whole module. Geometric identities are *exact* and are checked at 10^{-12}; comparisons against a numerical solver are checked at that solver's stated tolerance and no better — a check claiming 10^{-12} agreement with an interior point solver is a check that was not performed. Where a tolerance below is 10^{-9} rather than 10^{-12}, the looser number is deliberate and marks a quantity produced by an iterative solve. Keep one file per session, `s1.py` to `s7.py`, plus a shared `m1lib.py` that S7 imports.

One design decision, made now. Represent every convex set by a small object exposing three methods: `contains(x)`, `sample()` and `project(x)`. Represent a cone additionally by generators or by an inequality description, with a converter between them. Every validation check in S3 through S7 is then a loop over the same list of sets, and S4, S5 and S6 each cost a few lines instead of a rewrite. *This is the single decision that determines how long the module takes.*

4 Session map

Table 1: The seven sessions of M1, with the hour split and the deliverable of each. St, Wr and Cd are hours of studying this text, writing out proofs and exercises by hand, and code.

S	Title	St	Wr	Cd	Tot	Proof of concept
S1	Convex sets, hulls and Carathéodory	0.3	0.3	0.2	0.8	The Carathéodory reduction
S2	Relative interior, closure and closedness	0.3	0.3	0.2	0.8	The closedness failure, numerically
S3	Projection onto a closed convex set	0.4	0.3	0.5	1.2	The projection library
S4	Separation	0.3	0.5	0.2	1.0	`separate(C,x)` from `project`
S5	Cones, polars and Moreau's decomposition	0.3	0.3	0.2	0.8	Polars and the conic decomposition
S6	Closed cones and Farkas' lemma	0.5	0.5	0.4	1.4	The Farkas certificate
S7	Consolidation, capstone and checkpoint	0.2	0.3	0.5	1.0	The suite, plus the non-closed cone
	Total	**2.3**	**2.5**	**2.2**	**7.0**	

Table 2: Sources for a second view. *None of these is required.* Every proof this module uses is written out below; the column says what each book adds if you want another treatment after finishing a session.

Source	After	What it adds
Hiriart-Urruty and Lemaréchal, *Fundamentals of Convex Analysis*, part A	S1–S6	The same material with more examples and the infinite-dimensional caveats; A.3 and A.4 are the closest thing in print to S3 and S4
Boyd and Vandenberghe, ch. 2, §8.1	S1, S3–S5	A large catalogue of convex sets and of closed-form projections; the dual-cone conventions used in the engineering literature
Crouzeix and Ocaña, *Análisis convexo*	S1–S6	The MMC007 syllabus text, in Spanish; useful for fixing vocabulary in the language of the lectures
Rockafellar, *Convex Analysis*	S2, S4–S6	The reference statements later modules cite: §6, §11, §14, §17, §19, §22. Also the proofs omitted here (§6 and Thm. 11.3)
Bertsekas, *Nonlinear Programming*, App. B	S2, S6	Recession cones in far more detail than S2 needs, and the closedness criteria for images and sums
Schrijver, §7.3	S6	Farkas' variants collected, with the polyhedral rather than the separation proof — worth seeing once, as a contrast
Bertsimas and Tsitsiklis, §4.6	S6	Farkas read as an infeasibility certificate; this is the form M11 and M12 use
Beck, *First-Order Methods*, ch. 6	S3	A larger projection table, and the projection as a special proximal operator, which is M3
Farkas (1902); Gordan (1873); Moreau (1962)	S5, S6	The originals of Theorems 25, 27 and 20; Moreau is two pages

5 Setup and notation

The ambient space is $\mathbb{R}^n$ with the Euclidean inner product $\langle \cdot, \cdot \rangle$ and the norm $\|x\| = \langle x, x \rangle^{1/2}$. Everything here is finite-dimensional; the two places where that is essential rather than convenient are marked when they occur. Convex *functions* belong to M2 and are not used, with one exception flagged in S3.

Definition 1 (Convex set, hulls, cone)**.** $C \subseteq \mathbb{R}^n$ is *convex* if $(1-\lambda)x + \lambda y \in C$ whenever $x, y \in C$ and $\lambda \in [0,1]$. A *convex combination* of $x_1, \dots, x_k$ is $\sum \lambda_i x_i$ with $\lambda_i \geq 0$, $\sum \lambda_i = 1$; an *affine* combination drops $\lambda_i \geq 0$; a *conic* combination drops $\sum \lambda_i = 1$. $\operatorname{conv} S$, $\operatorname{aff} S$ and $\operatorname{cone} S$ denote the intersections of all convex, affine and convex-conic sets containing S. A set K is a *cone* if $\lambda x \in K$ for all $x \in K, \lambda \geq 0$; note $0 \in K$ for every cone here.

Definition 2 (Relative interior)**.** $\operatorname{ri} C$ is the interior of C relative to $\operatorname{aff} C$: the set of $x \in C$ for which there is $\varepsilon > 0$ with $B(x, \varepsilon) \cap \operatorname{aff} C \subseteq C$. The relative boundary is $\operatorname{cl} C \setminus \operatorname{ri} C$.

Definition 3 (Projection and distance)**.** For C nonempty closed convex, $P_C x = \arg\min_{c \in C} \|x - c\|$ and $\mathrm{d}_C(x) = \|x - P_C x\|$. Theorem 10 shows both are well defined.

Definition 4 (Polar and dual cone)**.** For $S \subseteq \mathbb{R}^n$, the *polar* is $S^\circ = \{y : \langle y, s \rangle \leq 0 \ \forall s \in S\}$ and the *dual* is $S^* = -S^\circ = \{y : \langle y, s \rangle \geq 0 \ \forall s \in S\}$. *This document uses the polar throughout.*

Two abbreviations used without comment: $\Delta_n = \{x \in \mathbb{R}^n : x \geq 0, \ \sum_i x_i = 1\}$ is the probability simplex, and for $A \in \mathbb{R}^{m \times n}$, $\operatorname{cone}(A) = \{Ax : x \geq 0\}$ is the cone generated by the columns of A.

Four warnings, all of which cause avoidable confusion later:

- **"Interior" is the wrong word.** The feasible set $\{x : Ax = b, \ x \geq 0\}$ of a linear program in standard form has empty interior in $\mathbb{R}^n$ whenever A has a nonzero row, and so does Δ_n. Every theorem in this plan that appears to require an interior point — Slater's condition in M4, the strictly feasible start of M13 — is stated with the relative interior of Definition 2. S2 exists to make that distinction reflexive.
- **Closedness is a hypothesis, not a technicality.** Strong separation is false without it; the sum of two closed convex sets need not be closed; and Farkas' lemma is an *alternative* — exactly one of two systems holds — rather than a mere implication only because $\operatorname{cone}(A)$ is closed, which is Lemma 24 and is the one genuinely nontrivial lemma of the module. The capstone in S7 exhibits what happens when it fails, and the answer is a nonzero duality gap.
- P_C **is not linear.** It is nonexpansive, which is weaker, and it is linear exactly when C is an affine subspace — the case MMC201's module M2 studies under the name "least squares". Expecting P_C to commute with a gradient step is the standard first error in reading the projected gradient method of M8, and S3 check 5 exists to make the non-linearity a number you have seen.
- **Three separations, three hypotheses.** *Proper* separation, *strong* separation (with a positive gap), and a *supporting* hyperplane (through a boundary point) are three different theorems. Which one is available decides whether the duality gap of M4 is zero. Do not use the word "separating" without saying which.

6 Session S1. Convex sets, hulls and Carathéodory

The question. The convex hull $\operatorname{conv} S$ is defined as an intersection over all convex sets containing S — an uncountable family — or, equivalently, as the set of *all* convex combinations of *finitely many* points of S, with no bound on how many. Neither description can be checked by a computer. If somebody claims $x \in \operatorname{conv} S$, what is the smallest amount of evidence that settles it?

The picture. Take a cloud of fifty points in the plane and a point x inside their hull. You never need fifty of them: find a triangle with vertices in the cloud that contains x, and three points suffice. The claim of this session is that "three" is "$n+1$" in $\mathbb{R}^n$, and that this is a dimension count rather than an accident. Here is the mechanism. If x is written using more than $n+1$ points, those points are affinely dependent. An affine dependence is a direction in weight-space along which the weights can be moved *without moving* x. Slide along it until some weight first hits zero. One point has been eliminated and x has not moved. Repeat.

That last paragraph is the entire proof. What follows writes it down.

Statements and proofs

Lemma 1 (The hull is the set of finite convex combinations)**.** $\operatorname{conv} S$ *equals the set* T *of all convex combinations of finitely many points of* S*.*

Proof. T is convex: a convex combination of two convex combinations is a convex combination. T contains S (take $k=1$). Hence $\operatorname{conv} S \subseteq T$.

For $T \subseteq \operatorname{conv} S$, let C be any convex set with $S \subseteq C$; we show $T \subseteq C$ by induction on the number k of atoms. For $k=1$ this is $S \subseteq C$. Suppose it holds for $k-1$ and let $x = \sum_{i=1}^{k} \lambda_i x_i$ with $x_i \in S$, $\lambda_i \geq 0$, $\sum \lambda_i = 1$. If $\lambda_k = 1$ then $x = x_k \in C$. Otherwise $\lambda_k < 1$, and

$$x = (1-\lambda_k)\, y + \lambda_k x_k, \qquad y = \sum_{i=1}^{k-1} \frac{\lambda_i}{1-\lambda_k}\, x_i.$$

The coefficients of y are nonnegative and sum to 1, so $y \in C$ by the inductive hypothesis, and then $x \in C$ by convexity. Since C was arbitrary, $T \subseteq \operatorname{conv} S$. □

Theorem 2 (Carathéodory)**.** *Let* $S \subseteq \mathbb{R}^n$ *and* $x \in \operatorname{conv} S$*. Then* x *is a convex combination of at most* $n+1$ *points of* S*.*

Proof. By Lemma 1, write $x = \sum_{i=1}^{k} \lambda_i x_i$ with $x_i \in S$, all $\lambda_i > 0$ (discard zero weights), $\sum \lambda_i = 1$, and k *minimal* among all such representations. Suppose, for contradiction, that $k \geq n+2$.

The $k-1 \geq n+1$ vectors $x_2 - x_1, \ldots, x_k - x_1$ lie in $\mathbb{R}^n$ and so are linearly dependent: there are $\mu_2, \ldots, \mu_k$, not all zero, with $\sum_{i=2}^{k} \mu_i (x_i - x_1) = 0$. Put $\mu_1 = -\sum_{i=2}^{k} \mu_i$. Then

$$\sum_{i=1}^{k} \mu_i x_i = 0, \qquad \sum_{i=1}^{k} \mu_i = 0, \qquad \mu \neq 0.$$

Because the μ_i sum to zero and are not all zero, at least one is strictly positive. Let

$$t^\star = \min\Big\{ \frac{\lambda_i}{\mu_i} \;:\; \mu_i > 0 \Big\},$$

attained at some index j, and set $\lambda'_i = \lambda_i - t^\star \mu_i$. Then:

- $\sum_i \lambda'_i = \sum_i \lambda_i - t^\star \sum_i \mu_i = 1$;
- $\sum_i \lambda'_i x_i = x - t^\star \cdot 0 = x$;
- $\lambda'_i \geq 0$: if $\mu_i > 0$ then $\lambda'_i = \mu_i(\lambda_i/\mu_i - t^\star) \geq 0$ by the definition of $t^\star$; if $\mu_i \leq 0$ then $\lambda'_i \geq \lambda_i > 0$ since $t^\star \geq 0$;

- $\lambda'_j = 0$ exactly.

So x is a convex combination of at most $k-1$ points of S, contradicting the minimality of k. Hence $k \le n+1$. □

Remark. Finite dimension is used once, in "$k-1 \ge n+1$ vectors in $\mathbb{R}^n$ are linearly dependent". In an infinite-dimensional space the theorem is false as stated; the correct substitute is a statement about closures, and it is one of the reasons infinite-dimensional convex analysis is a different subject.

Corollary 3 (Conic Carathéodory)**.** *If $x \in \operatorname{cone} S$ with $S \subseteq \mathbb{R}^n$, then $x = \sum_{i=1}^{k} \mu_i s_i$ with $\mu_i > 0$, $s_i \in S$ and $s_1, \dots, s_k$* linearly independent*; in particular $k \le n$.*

Proof. Take such a representation with k minimal and all $\mu_i > 0$. If $s_1, \dots, s_k$ were linearly dependent there would be $\nu \neq 0$ with $\sum_i \nu_i s_i = 0$; replacing ν by $-\nu$ if necessary, some $\nu_i > 0$. With $t^\star = \min\{\mu_i/\nu_i : \nu_i > 0\}$ attained at j, the coefficients $\mu_i - t^\star \nu_i$ are nonnegative, represent x, and vanish at j, contradicting minimality. So the s_i are independent and $k \le n$. □

Proposition 4. *If $K \subseteq \mathbb{R}^n$ is compact then* $\operatorname{conv} K$ *is compact.*

Proof. Let $\Phi : \Delta_{n+1} \times K^{n+1} \to \mathbb{R}^n$, $\Phi(\lambda, x_1, \dots, x_{n+1}) = \sum_{i=1}^{n+1} \lambda_i x_i$. It is continuous and its domain is compact, so its image is compact. The image is contained in $\operatorname{conv} K$; and it contains $\operatorname{conv} K$ by Theorem 2, since a combination of $k \le n+1$ points becomes one of exactly $n+1$ points by repeating a point with weight 0. Hence $\operatorname{conv} K$ is that image. □

Compactness cannot be weakened to closedness, and the following example is worth keeping: S2 explains what has to be added, S4 uses a relative of it, and S7 uses it again.

Example 5 (The convex hull of a closed set need not be closed)**.** Let $S = \{(u, v) : u > 0,\ v \ge 1/u\} \cup \{(0,0)\} \subseteq \mathbb{R}^2$. The first piece is closed (a sequence in it with $u_k \to 0^+$ has $v_k \to \infty$, so has no limit in $\mathbb{R}^2$), the second is a point, and a union of two closed sets is closed; so S is closed. But $\operatorname{conv} S$ is not. Indeed, for $\varepsilon \in (0, 1]$ take $u = \varepsilon^{-1/2}$ and the combination

$$(1 - \sqrt{\varepsilon})\,(0,0) + \sqrt{\varepsilon}\,\left(\varepsilon^{-1/2},\ \varepsilon^{1/2}\right) = (1, \varepsilon) \in \operatorname{conv} S,$$

so $(1, \varepsilon) \in \operatorname{conv} S$ for every $\varepsilon > 0$. Yet $(1, 0) \notin \operatorname{conv} S$: every point of S other than the origin has $v > 0$, so any convex combination with $v = 0$ must place all its weight on the origin and is therefore $(0, 0)$. Thus $(1, \varepsilon) \to (1, 0) \notin \operatorname{conv} S$.

Worked example

In $\mathbb{R}^2$ take $S = \{(1,0), (0,1), (-1,0), (0,-1)\}$ and $x = (0,0) = \frac{1}{4}\sum_i s_i$. This uses $k = 4 > n+1 = 3$ atoms, so Theorem 2 says it can be reduced. Run the proof.

The four points are affinely dependent: $(1,0) - (0,1) + (-1,0) - (0,-1) = 0$ with coefficients $\mu = (1, -1, 1, -1)$ summing to 0. The positive entries are $\mu_1 = \mu_3 = 1$, and

$$t^\star = \min\{\lambda_1/\mu_1,\ \lambda_3/\mu_3\} = \min\{\tfrac{1}{4}, \tfrac{1}{4}\} = \tfrac{1}{4}.$$

Then $\lambda' = \lambda - t^\star \mu = (0, \frac{1}{2}, 0, \frac{1}{2})$, giving $x = \frac{1}{2}(0,1) + \frac{1}{2}(0,-1)$: two atoms, and indeed $2 \le 3$. Note that *two* weights vanished at once here, because the minimum in $t^\star$ was attained twice; the proof only promises one.

Where this is used later

Corollary 3 is the engine of Lemma 24 in S6, which is the hypothesis of Farkas' lemma, which is the theorem the rest of the plan quotes most. The module's logical chain runs through its shortest session. Theorem 2 also bounds the support of a Frank–Wolfe iterate in M8 and explains the vertex solutions of M11.

Reinforcement learning. A stochastic policy at one state is a point of $\Delta_{|\mathcal{A}|} = \operatorname{conv}\{\delta_a\}$, so Theorem 2 there says only that a distribution is a mixture of point masses. The content appears one level up. The set of discounted occupancy measures of a finite Markov decision process is a polytope in $\mathbb{R}^{|\mathcal{S}||\mathcal{A}|}$ whose vertices are the deterministic policies, so *every achievable occupancy measure — hence every achievable behaviour — is a mixture of at most* $|\mathcal{S}||\mathcal{A}| + 1$ *deterministic policies.* That finiteness is why M11's linear program has a vertex solution and why one may search over deterministic policies without loss.

Proof of concept: the Carathéodory reduction

Implement `caratheodory(x, S)` by iterating the elimination step of the proof: solve for an affine dependence (a null vector of the matrix with rows $x_i^\top$ and a row of ones), compute $t^\star$, update, drop the vanished atom. Implement the set objects for the box, the ball, the simplex and a random polytope, each with `contains` and `sample`; `project` arrives in S3.

Validation.

1. On 10^3 random instances with $n \in \{2, 5, 10\}$ and $|S| = 50$: the output has at most $n + 1$ atoms, weights are nonnegative and sum to 1 within 10^{-12}, and the combination reconstructs x within 10^{-12}.
2. The reduction is exact, not approximate: assert that at each step the eliminated weight is *exactly* 0, by construction rather than by tolerance. If this fails, $t^\star$ was taken as a minimum over the wrong index set — the minimum runs over $\mu_i > 0$ only.
3. Reproduce the worked example above and confirm the routine returns two atoms with weights $\frac{1}{2}, \frac{1}{2}$, not three.
4. Feed a point already supported on $k \le n+1$ atoms and confirm the routine returns it unchanged. This catches the common bug of eliminating unconditionally.

Deliverable. `s1.py`, and `caratheodory` plus the set objects in `m1lib.py`.

Exercises

E1.1 Reproduce the proof of Theorem 2 with the document closed. Mark the one sentence that uses finite dimension.

E1.2 Show that an arbitrary intersection of convex sets is convex, and give two convex sets whose union is not. Deduce that the feasible set of an optimization problem is convex as soon as each constraint set is — and note that this closure property is the entire practical reason convex programming is a usable class of problems.

E1.3 Verify every claim in Example 5, including the arithmetic of the displayed combination.

E1.4 Prove Corollary 3 without looking, and show $\operatorname{cone}\{a_1, \dots, a_m\} = \{Ax : x \ge 0\}$ where A has the a_i as columns. **[Reused in S6: this corollary is the proof of Lemma 24.]**

Session gate. Checks 1–4 pass and E1.4 is written out.

7 Session S2. Relative interior, closure and the closedness hypothesis

The question. Every theorem in the rest of this module says "closed convex". It is tempting to read "closed" as a technical courtesy. Is it? And separately: optimization constantly produces sets with *empty interior* — anything cut out by an equality constraint — so what replaces "interior point" in the hypotheses of the theorems that seem to need one?

The picture. Two pictures, one for each half of the question.

First: the probability simplex in $\mathbb{R}^3$ is a solid triangle sitting inside a plane. It has no interior point in $\mathbb{R}^3$ — every ball around any of its points sticks out of the plane. But inside the plane it plainly has an inside. The fix is to measure interiority relative to the affine hull, and it is not a cosmetic fix: the entire hypothesis of the strong duality theorem in M4 is of this form.

Second: take the region above a hyperbola in the right half-plane, and the vertical axis. Both are closed and convex. Add them. Going far to the right along the hyperbola, the region's points come arbitrarily close to the horizontal axis; sliding those points vertically by any amount — which is what adding the axis does — sweeps out the entire *open* right half-plane. The sum has lost its boundary, because the two sets escape to infinity in complementary directions and the sum accumulates on points that neither sum ever reaches. This is not pathology. It is the mechanism of every duality gap in the plan.

Statements and proofs

Proposition 5. *If $C \subseteq \mathbb{R}^n$ is convex and nonempty then* $\operatorname{ri} C \neq \emptyset$.

Proof. Let $d = \dim \operatorname{aff} C$ and choose $x_0, \dots, x_d \in C$ affinely independent (possible by the definition of d). The map $\psi(\lambda_1, \dots, \lambda_d) = x_0 + \sum_{i=1}^{d} \lambda_i (x_i - x_0)$ is an affine bijection from $\mathbb{R}^d$ onto $\operatorname{aff} C$, and it is a homeomorphism because the $x_i - x_0$ are linearly independent. The point $b = \psi(\frac{1}{d+1}, \dots, \frac{1}{d+1}) = \frac{1}{d+1} \sum_{i=0}^{d} x_i$ is the barycentre of the simplex $\operatorname{conv}\{x_0, \dots, x_d\} \subseteq C$. Its preimage lies in the open set $\{\lambda : \lambda_i > 0,\ \sum \lambda_i < 1\}$, whose image under ψ consists of convex combinations of the x_i with all weights positive, hence lies in C. As ψ is a homeomorphism onto $\operatorname{aff} C$, that image is relatively open, so $b \in \operatorname{ri} C$. □

Proposition 6 (Line segment principle)**.** *If $x \in \operatorname{ri} C$, $y \in \operatorname{cl} C$ and $\lambda \in [0, 1)$, then $z = (1-\lambda)x + \lambda y \in \operatorname{ri} C$.*

Proof. Work inside $\operatorname{aff} C$, so we may assume $\operatorname{aff} C = \mathbb{R}^n$ and ri = int. Let $\varepsilon > 0$ with $B(x, \varepsilon) \subseteq C$, and set $r = (1-\lambda)\varepsilon/2 > 0$. Let w satisfy $\|w - z\| < r$. Since $y \in \operatorname{cl} C$ there is $y' \in C$ with $\|y - y'\| < r$. Define

$$x' = \frac{w - \lambda y'}{1 - \lambda}, \qquad \text{so that} \qquad w = (1-\lambda)x' + \lambda y'.$$

Then, using $z = (1-\lambda)x + \lambda y$,

$$\|x' - x\| = \frac{\|w - \lambda y' - (1-\lambda)x\|}{1-\lambda} = \frac{\|(w - z) + \lambda(y - y')\|}{1-\lambda} \leq \frac{r + \lambda r}{1-\lambda} \leq \frac{2r}{1-\lambda} = \varepsilon.$$

Hence $x' \in B(x, \varepsilon) \subseteq C$, and $w = (1-\lambda)x' + \lambda y' \in C$ by convexity. So $B(z, r) \subseteq C$ and $z \in \operatorname{ri} C$. □

Proposition 7. *For C convex nonempty: (1)* $\operatorname{cl}(\operatorname{ri} C) = \operatorname{cl} C$*; (2)* $\operatorname{ri}(\operatorname{cl} C) = \operatorname{ri} C$.

Proof. (1) $\subseteq$ is immediate. For $\supseteq$, take $y \in \operatorname{cl} C$ and, by Proposition 5, some $x \in \operatorname{ri} C$. By Proposition 6, $(1-\lambda)x + \lambda y \in \operatorname{ri} C$ for every $\lambda < 1$; letting $\lambda \to 1^-$ exhibits y as a limit of points of $\operatorname{ri} C$. Statement (2) is proved similarly and is stated here without proof — it is used only for orientation, and its proof adds nothing this module needs. □

Example 6 (The simplex)**.** $\Delta_n \subseteq H = \{x : \sum_i x_i = 1\}$, a hyperplane, which has empty interior in $\mathbb{R}^n$; hence $\operatorname{int} \Delta_n = \emptyset$. On the other hand $\operatorname{aff} \Delta_n = H$, and $\operatorname{ri} \Delta_n = \{x \in H : x_i > 0 \ \forall i\}$: if all $x_i > 0$ then any $y \in H$ with $\|y - x\| < \min_i x_i$ has $y_i > 0$, so $y \in \Delta_n$; and if some $x_j = 0$, then $x + \delta(e_j - e_k) \in H$ for $k \neq j$ has a negative j-th coordinate for every $\delta < 0$ arbitrarily small in norm, so $x \notin \operatorname{ri} \Delta_n$.

Example 7 (A sum of closed convex sets that is not closed)**.** Let $C = \{(u, v) : u > 0,\ v \geq 1/u\}$ and $D = \{0\} \times \mathbb{R}$. Both are convex, and both are closed (C by the argument of Example 5). Their sum is

$$C + D = \{(u, v + t) : u > 0,\ v \geq 1/u,\ t \in \mathbb{R}\} = \{(u, s) : u > 0,\ s \in \mathbb{R}\},$$

the *open* right half-plane, which is not closed: $(1/k, 0) \in C + D$ for every k, and the limit $(0, 0)$ is not in $C + D$. The same set shows that the affine image of a closed convex set need not be closed, since the projection of C onto its first coordinate is $(0, \infty)$.

Why does this happen, and when does it not? The answer is about directions of escape.

Definition 8 (Recession cone)**.** For C closed convex nonempty, $0^+C = \{d : x + td \in C \ \forall x \in C,\ t \geq 0\}$.

Lemma 8. *Let C be closed convex, $c_k \in C$ with $\|c_k\| \to \infty$ and $c_k/\|c_k\| \to u$. Then $u \in 0^+C$.*

Proof. Fix $x \in C$ and $t \geq 0$. For k large, $t/\|c_k\| \leq 1$, so by convexity $x + \frac{t}{\|c_k\|}(c_k - x) \in C$. As $k \to \infty$ this converges to $x + tu$, and C is closed, so $x + tu \in C$. □

Proposition 9 (Recession criterion)**.** *Let C, D be closed convex nonempty with $0^+C \cap (-0^+D) = \{0\}$ — in particular, with either set bounded. Then $C + D$ is closed.*

Proof. Let $z_k = c_k + d_k \to z$ with $c_k \in C$, $d_k \in D$. If (c_k) has a bounded subsequence, pass to it and to a convergent sub-subsequence $c_k \to c \in C$ (closed); then $d_k = z_k - c_k \to z - c$, which lies in D (closed), so $z \in C + D$. Otherwise $\|c_k\| \to \infty$, and since (z_k) is bounded, $\|d_k\| \to \infty$ with $\|d_k\|/\|c_k\| \to 1$. Pass to a subsequence with $c_k/\|c_k\| \to u$, $\|u\| = 1$; then $d_k/\|d_k\| = \frac{\|c_k\|}{\|d_k\|} \cdot \frac{z_k - c_k}{\|c_k\|} \to -u$. By Lemma 8, $u \in 0^+C$ and $-u \in 0^+D$, so $u \in 0^+C \cap (-0^+D) = \{0\}$, contradicting $\|u\| = 1$. (If either set is bounded its recession cone is $\{0\}$ and the hypothesis holds automatically.) □

In Example 7, $0^+C = \mathbb{R}^2_+$ and $0^+D = -0^+D = \{0\} \times \mathbb{R}$, so $0^+C \cap (-0^+D) = \{0\} \times \mathbb{R}_+ \neq \{0\}$: the hypothesis fails exactly where the closedness does, and the offending direction $(0, 1)$ is visibly the direction in which the two sets escape.

Worked example

Take C and D of Example 7 and ask for the pair (c, d) minimizing $\|c+d\|$. Parameterize $c = (u, 1/u)$ with $u > 0$ and $d = (0, t)$; then $\|c + d\|^2 = u^2 + (1/u + t)^2$, and for each u the best t is $t = -1/u$, giving $\|c + d\| = u$. So the infimum over the parameterization is $\inf_{u>0} u = 0$, *and it is attained at no $u > 0$*: the minimizing sequence is $u_k \to 0^+$, at which $\|d_k\| = 1/u_k \to \infty$. The infimum 0 is the distance from the origin to $C + D$, the origin is in $\operatorname{cl}(C + D)$, and the minimizer has escaped to infinity in the direction $(0, -1) \in -0^+D$ that Proposition 9 forbids. *An infimum approached only at infinity is what "not closed" means operationally*, and this is the phrase to remember.

Now intersect C with the ball $\bar{B}(0,R)$, which makes it compact. The constraint $u^2+1/u^2 \le R^2$ forces $u \ge u_{\min}(R) > 0$, so the minimum becomes $u_{\min}(R) > 0$ and is attained on the sphere. Compactness restores attainment and the optimal value is strictly positive for every finite R, which is Proposition 9 seen from the other side.

Where this is used later

Every "closed convex" in S3–S6 is this session. Concretely: Theorem 10 needs closedness for existence; Theorem 17 needs Proposition 9 to know $C - D$ is closed; Lemma 24 is a closedness statement and is the hypothesis of Farkas. In M4, the duality gap is nonzero exactly when a certain perturbation set fails to be closed, and the mechanism is the one in Example 7.

Reinforcement learning. The set of discounted occupancy measures of a finite Markov decision process is a polytope, hence compact, so by Proposition 4 and Weierstrass the supremum of expected return over policies is attained. M11 may therefore write max rather than sup, and an optimal policy exists — a geometric fact, not a probabilistic one.

Proof of concept: the closedness failure, numerically

Implement the parameterized minimization of the worked example on a grid over u, refined five times, with the optimal $t = -1/u$ substituted in closed form.

Validation.

1. Tabulate, at five successive refinements, the best value $\|c+d\|$ and the minimizing u and $\|d\| = 1/u$. Confirm the value decreases toward 0 while $\|d\|$ roughly doubles at each halving of the grid tolerance. *Print the table.*
2. Repeat with $C \cap \bar{B}(0,R)$ for $R \in \{2, 10, 100\}$: confirm the minimum is now attained at an interior grid point, that the minimizer satisfies $\|c\| = R$ to the grid resolution, and that the optimal value is strictly positive and decreasing in R. *This tests Proposition 9 as an if-and-only-if rather than illustrating it.*
3. Example 6 numerically for $n = 3$: confirm by sampling that no point of Δ_3 has a full-dimensional ball inside it — every random perturbation in $\mathbb{R}^3$ leaves the affine hull — while for each sampled x with all $x_i > 0$ you can exhibit an explicit ε such that every y in aff Δ_3 with $\|y - x\| < \varepsilon$ lies in Δ_3. Report the empirical relation between ε and $\min_i x_i$; the proof of Example 6 predicts $\varepsilon = \min_i x_i$.

Deliverable. `s2.py` and the divergence table of check 1.

Exercises

E2.1 Reproduce Proposition 6 with the document closed, and deduce Proposition 7(1).

E2.2 Verify Example 6 in full, and state the consequence for how Slater's condition must be written. **[Reused in M4.]**

E2.3 Verify Example 7, compute both recession cones, and identify the direction in which Proposition 9 fails.

E2.4 Prove that the image of a *polyhedron* $\{x : Ax \le b\}$ under a linear map is a polyhedron, hence closed, so that no counterexample of Example 7's kind can be polyhedral. The argument is Fourier–Motzkin elimination: eliminate one variable at a time by pairing each inequality in which it has a positive coefficient with each in which it has a negative one. **[Reused in M14: projecting out the complicating variables is exactly this operation, and it is why Benders' master problem is a linear program.]**

Session gate. Checks 1 and 2 pass; E2.2 and E2.4 are written out.

8 Session S3. Projection onto a closed convex set

The question. Given a point x and a set C, is there a nearest point of C to x? Is it unique? And — the question that actually matters — is there a way to *recognize* the nearest point, some equation or inequality it satisfies and no other point of C does? Without such a characterization the nearest point is a definition and nothing follows from it.

The picture. For a subspace you know the answer: drop a perpendicular, and the characterization is $x - p \perp C$. For a general convex set the perpendicular cannot be exact, because C has a boundary and p may sit on it; the equality must weaken to an inequality. Stand at x and look at p, the nearest point. If some $y \in C$ were at an angle strictly less than $90°$ from the direction $x - p$, you could walk a little way from p towards y — staying in C, by convexity — and get closer to x. So no $y \in C$ is at an angle below $90°$:

$$\langle x - p,\, y - p\rangle \le 0 \qquad \text{for every } y \in C.$$

That inequality is the whole session. Everything else — existence, uniqueness, nonexpansiveness, every closed formula, the separating hyperplane of S4, Moreau's decomposition in S5, the Farkas certificate of S6 — is obtained by using it.

Statements and proofs

Theorem 10 (Projection theorem)**.** *Let $C \subseteq \mathbb{R}^n$ be nonempty, closed and convex, and let $x, z \in \mathbb{R}^n$.*

1. There is exactly one point $P_C x \in C$ minimizing $\|x - c\|$ over $c \in C$.
2. For $p \in C$: $p = P_C x$ if and only if $\langle x - p,\, y - p\rangle \le 0$ for every $y \in C$.
3. $\|P_C x - P_C z\|^2 \le \langle x - z,\, P_C x - P_C z\rangle$; consequently $\|P_C x - P_C z\| \le \|x - z\|$.

Proof. (1) Existence. Fix any $c_0 \in C$ and put $R = \|x - c_0\|$. The set $C \cap \bar{B}(x, R)$ is nonempty (it contains c_0), bounded, and closed — *this is the only place closedness of C is used* — hence compact. The continuous function $c \mapsto \|x - c\|$ attains its minimum on it, at some p. Any $c \in C$ outside the ball has $\|x - c\| > R \ge \|x - p\|$, so p minimizes over all of C.

Uniqueness. Let $p, q \in C$ both attain the minimum value δ. Their midpoint $m = (p + q)/2$ lies in C by convexity — *this is where convexity is used* — and the parallelogram identity gives

$$\|x - p\|^2 + \|x - q\|^2 = 2\|x - m\|^2 + \tfrac{1}{2}\|p - q\|^2,$$

that is, $2\delta^2 = 2\|x - m\|^2 + \frac{1}{2}\|p - q\|^2$. Since $\|x - m\| \ge \delta$, we get $\frac{1}{2}\|p - q\|^2 \le 0$, so $p = q$.

(2) ($\Leftarrow$). Suppose $p \in C$ satisfies the inequality. For any $y \in C$,

$$\|x - y\|^2 = \|(x - p) - (y - p)\|^2 = \|x - p\|^2 - 2\langle x - p,\, y - p\rangle + \|y - p\|^2 \;\ge\; \|x - p\|^2,$$

so p is a minimizer, and by (1) it is *the* minimizer.

($\Rightarrow$). Let $p = P_C x$ and $y \in C$. For $t \in (0, 1]$ the point $p + t(y - p)$ lies in C by convexity, so

$$\|x - p - t(y - p)\|^2 \;\ge\; \|x - p\|^2 \quad\Longrightarrow\quad -2t\langle x - p,\, y - p\rangle + t^2\|y - p\|^2 \;\ge\; 0.$$

Divide by $t > 0$ to get $-2\langle x - p,\, y - p\rangle + t\|y - p\|^2 \ge 0$, and let $t \downarrow 0$: $\langle x - p,\, y - p\rangle \le 0$.

(3). Write $p = P_C x$, $q = P_C z$. Applying (2) at x with $y = q$, and at z with $y = p$:

$$\langle x - p,\, q - p\rangle \le 0, \qquad \langle z - q,\, p - q\rangle \le 0.$$

Adding them and writing $\langle z - q, p - q\rangle = \langle q - z, q - p\rangle$ gives $\langle (x - z) - (p - q),\, q - p\rangle \le 0$, i.e. $-\langle x - z,\, p - q\rangle + \|p - q\|^2 \le 0$, which is the claim. Cauchy–Schwarz then gives $\|p - q\|^2 \le \|x - z\|\|p - q\|$, hence $\|p - q\| \le \|x - z\|$. □

Remark. Part (3) is called *firm* nonexpansiveness and is strictly stronger than the nonexpansiveness deduced from it. M9 needs the firm version: it is what makes proximal iterates Fejér monotone with respect to the solution set, and plain nonexpansiveness is not enough for that. Notice also that (3) was proved by using (2) twice and adding — a manoeuvre that returns in S5.

Proposition 11. d_C *is convex and* 1*-Lipschitz, and* $\varphi = \frac{1}{2}\,\mathrm{d}_C^2$ *is differentiable everywhere with* $\nabla\varphi(x) = x - P_C x$.

Proof. Convexity. For x, y and $\lambda \in [0,1]$, the point $\lambda P_C x + (1-\lambda)P_C y$ lies in C, so

$$\mathrm{d}_C(\lambda x + (1-\lambda)y) \le \|\lambda(x - P_C x) + (1-\lambda)(y - P_C y)\| \le \lambda\,\mathrm{d}_C(x) + (1-\lambda)\,\mathrm{d}_C(y).$$

Lipschitz. $\mathrm{d}_C(x) \le \|x - P_C z\| \le \|x - z\| + \mathrm{d}_C(z)$, and symmetrically.

Differentiability. Put $g = x - P_C x$. Using $P_C x \in C$ as a competitor for y,

$$\varphi(y) \le \tfrac{1}{2}\|y - P_C x\|^2 = \varphi(x) + \langle g,\, y - x\rangle + \tfrac{1}{2}\|y - x\|^2,$$

which bounds the error above by $\frac{1}{2}\|y-x\|^2$. Using $P_C y$ as a competitor at x,

$$\varphi(x) \le \tfrac{1}{2}\|x - P_C y\|^2 = \varphi(y) + \langle y - P_C y,\, x - y\rangle + \tfrac{1}{2}\|x - y\|^2,$$

that is, $\varphi(y) \ge \varphi(x) + \langle y - P_C y,\, y - x\rangle - \frac{1}{2}\|y-x\|^2$. By Theorem 10(3) the map $y \mapsto y - P_C y$ is continuous, so $\langle y - P_C y,\, y - x\rangle = \langle g,\, y - x\rangle + o(\|y-x\|)$. Combining the two bounds, $\varphi(y) = \varphi(x) + \langle g,\, y - x\rangle + o(\|y-x\|)$. □

Remark. Proposition 11 is the one place this module touches convex functions. It is included because M3 will recognize φ as the *Moreau envelope* of the indicator function of C and P_C as its *proximal operator*, and will prove the gradient formula for a general convex function in place of an indicator. The proof above is that general proof, specialized.

Now the closed forms. Each is verified the same way: guess p, then check the inequality of Theorem 10(2). *No optimization is needed once the characterization is in hand*, and this is the practical payoff of part (2).

Proposition 12 (Three easy projections)**.** *1. Halfspace* $C = \{y : \langle a,\, y\rangle \le \beta\}$, $a \neq 0$*:* $P_C x = x - \dfrac{(\langle a,\, x\rangle - \beta)_+}{\|a\|^2}\, a$.

2. Affine set $C = \{y : Ay = b\}$ *with* A *of full row rank:* $P_C x = x - A^\top(AA^\top)^{-1}(Ax - b)$.

3. Ball $C = \bar{B}(c, r)$*:* $P_C x = x$ *if* $\|x - c\| \le r$*, and* $c + r\frac{x-c}{\|x-c\|}$ *otherwise. Box* $C = \prod_i [l_i, u_i]$*: componentwise clipping.*

Proof. (1) If $\langle a,\, x\rangle \le \beta$ then $x \in C$ and $p = x$ trivially. Otherwise let $p = x - \theta a$ with $\theta = (\langle a,\, x\rangle - \beta)/\|a\|^2 > 0$; then $\langle a,\, p\rangle = \beta$ so $p \in C$, and for $y \in C$, $\langle x - p,\, y - p\rangle = \theta\langle a,\, y - p\rangle = \theta(\langle a,\, y\rangle - \beta) \le 0$. Apply Theorem 10(2). (2) With $p = x - A^\top(AA^\top)^{-1}(Ax - b)$ one checks $Ap = b$, so $p \in C$; and for $y \in C$, $x - p = A^\top\mu$ with $\mu = (AA^\top)^{-1}(Ax - b)$, so $\langle x - p,\, y - p\rangle = \langle \mu,\, Ay - Ap\rangle = \langle \mu,\, b - b\rangle = 0 \le 0$. *The inequality holds with equality, which is exactly the statement that for an affine set the characterization degenerates to orthogonality* — the normal equations of least squares. (3) Analogous; both are left as E3.2. □

The simplex is the one that matters for the rest of the plan, and it is also the one where the obvious guess is wrong.

Proposition 13 (Projection onto the probability simplex)**.** *Let* $x \in \mathbb{R}^n$ *and let* θ *be the unique solution of* $\sum_{i=1}^n (x_i - \theta)_+ = 1$*. Then* $P_{\Delta_n}(x)_i = (x_i - \theta)_+$.

Proof. θ exists and is unique. $g(\theta) = \sum_i (x_i - \theta)_+$ is continuous and nonincreasing, $g(\theta) = 0$ for $\theta \geq \max_i x_i$, and $g(\theta) \to +\infty$ as $\theta \to -\infty$; moreover g is strictly decreasing wherever it is positive. Hence $g(\theta) = 1$ has exactly one solution.

The formula. Put $p_i = (x_i - \theta)_+$. Then $p \geq 0$ and $\sum_i p_i = g(\theta) = 1$, so $p \in \Delta_n$. Observe

$$x_i - p_i = \begin{cases} \theta, & x_i \geq \theta \ \text{ (i.e. } p_i = x_i - \theta \geq 0), \\ x_i < \theta, & x_i < \theta \ \text{ (i.e. } p_i = 0), \end{cases} \qquad \text{so in all cases } x_i - p_i \leq \theta,$$

with equality whenever $p_i > 0$. Now take any $y \in \Delta_n$:

$$\langle x - p,\, y - p \rangle = \sum_i (x_i - p_i) y_i - \sum_i (x_i - p_i) p_i.$$

In the second sum only indices with $p_i > 0$ contribute, and there $x_i - p_i = \theta$, so that sum equals $\theta \sum_i p_i = \theta$. In the first sum, $y_i \geq 0$ and $x_i - p_i \leq \theta$, so it is at most $\theta \sum_i y_i = \theta$. Hence $\langle x - p,\, y - p \rangle \leq \theta - \theta = 0$, and Theorem 10(2) identifies p as the projection. □

Remark. The formula subtracts a *common threshold* θ and then clips. The map that clips first and renormalizes, $x \mapsto (x)_+ / \sum_i (x_i)_+$, is a different map, and the worked example below shows how different. Sorting x makes θ computable in $O(n \log n)$: with $u_1 \geq \cdots \geq u_n$ the sorted entries and $\rho = \max\{j : u_j - \frac{1}{j}(\sum_{i \leq j} u_i - 1) > 0\}$, one has $\theta = \frac{1}{\rho}(\sum_{i \leq \rho} u_i - 1)$.

Worked example

Let $x = (0.5,\ 0.6,\ -0.2,\ 1.1) \in \mathbb{R}^4$. Sorted descending, $u = (1.1, 0.6, 0.5, -0.2)$ with partial sums $1.1,\ 1.7,\ 2.2,\ 2.0$. Test the condition $u_j - \frac{1}{j}(\sum_{i \leq j} u_i - 1) > 0$:

$$\begin{aligned} j = 1 &: \ 1.1 - 0.1 = 1.0 > 0, & \qquad j = 2 &: \ 0.6 - 0.35 = 0.25 > 0, \\ j = 3 &: \ 0.5 - 0.4 = 0.1 > 0, & \qquad j = 4 &: \ -0.2 - 0.25 = -0.45 < 0. \end{aligned}$$

So $\rho = 3$ and $\theta = \frac{1}{3}(2.2 - 1) = 0.4$. The projection is

$$P_\Delta(x) = \big((0.5 - 0.4)_+,\ (0.6 - 0.4)_+,\ (-0.2 - 0.4)_+,\ (1.1 - 0.4)_+\big) = (0.1,\ 0.2,\ 0,\ 0.7),$$

which is nonnegative and sums to 1. Check the theorem: $x - p = (0.4, 0.4, -0.2, 0.4)$, and indeed every entry is $\leq \theta = 0.4$, with equality exactly at the coordinates where $p_i > 0$.

Now the naive map: $(x)_+ = (0.5, 0.6, 0, 1.1)$ with sum 2.2, giving $(0.2273,\ 0.2727,\ 0,\ 0.5)$. The two answers differ by

$$\|(0.1, 0.2, 0, 0.7) - (0.2273, 0.2727, 0, 0.5)\| = \|(-0.1273,\ -0.0727,\ 0,\ 0.2)\| \approx 0.248,$$

which is a quarter of the diameter of the simplex — not a rounding difference. *The naive map is not the projection, and now you have seen the number.* Note also where they differ in kind: the projection preserves the *gaps* $x_i - x_j$ among surviving coordinates, while renormalization preserves their *ratios*.

Where this is used later

P_C is not a computational convenience introduced here for later use. It is the separating vector of S4, written $x - P_C x$; the stationarity condition of M8, written $x = P_C(x - t\nabla f(x))$; the Moreau decomposition of S5; and the Farkas certificate of S6. With C an affine subspace, Theorem 10(2) becomes the orthogonality $\langle x - p,\, y - p \rangle = 0$ of Proposition 12(2), which is the normal equations, which is the whole of MMC201's module M2. One theorem, four appearances.

Reinforcement learning. P_Δ is the operation every projected policy-gradient step performs, and Proposition 13 is why it costs $O(n \log n)$ per state rather than a quadratic program. Read what the formula says: it zeroes every action whose score falls below a common threshold. *A projected policy step is sparsifying*, and can reach — and stop at — a deterministic policy in finitely many steps, whereas a softmax parameterization assigns positive probability to every action forever. That contrast between the Euclidean and the entropic geometry is the subject of M9, and the clip-and-renormalize error above is the standard way it gets silently broken in code.

Proof of concept: the projection library

Implement `project` for the halfspace, the affine set $\{y : Ay = b\}$, the ball, the box, the second-order cone $K = \{(z, t) : \|z\| \le t\}$ and Δ_n, attaching each to the set objects of S1. For the second-order cone the formula is: (z, t) if $\|z\| \le t$; 0 if $\|z\| \le -t$; and otherwise $\frac{\|z\|+t}{2\|z\|}(z, \|z\|)$ — derive it in E3.4 rather than trusting it.

Validation.

1. The characterization: for each set, 10^3 random exterior x and 10^4 sampled $y \in C$, $\max\langle x - P_C x,\, y - P_C x\rangle \le 10^{-12}$.
2. Firm nonexpansiveness and nonexpansiveness on 10^4 random pairs: report the maximum violation of each (both $\le 10^{-12}$), and the maximum ratio $\|P_C x - P_C z\|/\|x - z\|$, which must not exceed 1 and should be found near 1 for pairs already in C.
3. Each closed form agrees with a `cvxpy` solve of $\min\|y - x\|^2$ s.t. $y \in C$ to 10^{-8} — the solver's tolerance, not machine precision — on 500 random instances per set.
4. Proposition 13: the output lies in Δ_n to 10^{-14}; and timing over $n = 10^2, \ldots, 10^6$ gives a fitted slope of log(time) against $\log n$ of $1 + o(1)$, not 2. *Report the fitted slope.*
5. P_C **is not linear.** Exhibit C, x, z, λ with $P_C(\lambda x + (1-\lambda)z) \ne \lambda P_C x + (1-\lambda)P_C z$, and report the gap. Then verify that for C an affine set the same expression vanishes to 10^{-12}. *Both halves are required; the second is what makes the first surprising.*
6. **Reproduce the worked example.** On the x above, confirm $\theta = 0.4$ and $p = (0.1, 0.2, 0, 0.7)$ exactly, and that the clip-and-renormalize map returns $(0.2273, 0.2727, 0, 0.5)$ at distance 0.248 from it. Then report the maximum discrepancy between the two maps over 10^4 random x in $\mathbb{R}^{50}$.

Deliverable. `s3.py` and `project` in `m1lib.py`. **Keep it:** S4 builds separating hyperplanes from it, S5 builds Moreau's decomposition from it, S6's Farkas certificate is one of its outputs, and M8 and M9 import it unchanged.

Exercises

E3.1 Reproduce Theorem 10, all three parts, with the document closed. Mark the one line that uses closedness and the one line that uses convexity.

E3.2 Prove Proposition 12(3), and verify that in part (2) the characterizing inequality holds with equality. Then write the normal equations of least squares and identify them with that equality. **[This is MMC201's M2.]**

E3.3 Prove Proposition 13 without looking, and state precisely where $y \in \Delta_n$ (rather than merely $y \ge 0$) was used.

E3.4 Derive the second-order cone projection stated above from Theorem 10(2). Hint: by symmetry the answer has the form $\alpha(z, \|z\|)$ on the "otherwise" branch; determine α from the characterization.

E3.5 Prove Proposition 11. Where does convexity of C enter, and what fails for d_C when C is closed but not convex?

Session gate. Checks 1, 2, 3, 5 and 6 pass, and E3.1 is written out in full.

9 Session S4. Separation

The question. A convex set has two descriptions. The *primal* one lists its points: $C = \operatorname{conv} S$, or $C = \{Ax : x \geq 0\}$. The *dual* one lists the constraints its points satisfy: $C = \{y : \langle a,\, y\rangle \leq \alpha \text{ for every } (a, \alpha) \in \mathcal{H}\}$. Every duality in this course is a passage between the two. But why should the second description exist at all, and how would one produce a constraint that a given exterior point violates?

The picture. Suppose $x \notin C$ with C closed convex. Project: let $p = P_C x$ and $a = x - p \neq 0$. The vector a points from the set towards x. By the inequality of S3, no point of C lies at an angle below 90° from a as seen from p; so along the direction a, the whole of C sits at or behind the level of p, while x sits $\|a\|^2$ beyond it. The hyperplane through the midpoint of the segment $[p, x]$, perpendicular to a, separates them — and the gap is not merely positive, it is exactly $\|a\|^2 = \mathrm{d}_C(x)^2$. *The separating hyperplane is not an existence statement. It is the output of the projection routine you wrote in S3.*

That construction requires closedness, and when closedness fails the gap collapses to zero: the sets can still be separated, but only by a hyperplane that touches both. Keeping those two cases apart is what the three separation theorems do.

Statements and proofs

Theorem 14 (Strong separation of a point)**.** *Let $C \subseteq \mathbb{R}^n$ be nonempty closed convex and $x \notin C$. Put $a = x - P_C x \neq 0$. Then*

$$\langle a,\, c\rangle \;\leq\; \langle a,\, x\rangle - \|a\|^2 \qquad \text{for every } c \in C.$$

Hence with $\alpha = \langle a,\, x\rangle - \|a\|^2$ we have $\sup_{c\in C}\langle a,\, c\rangle \leq \alpha < \langle a,\, x\rangle$, and the gap $\langle a,\, x\rangle - \alpha$ equals $\|a\|^2 = \mathrm{d}_C(x)^2$.

Proof. Write $p = P_C x$, so $a = x - p$ and $a \neq 0$ because $x \notin C$. Theorem 10(2) says $\langle a,\, c - p\rangle \leq 0$ for every $c \in C$, i.e. $\langle a,\, c\rangle \leq \langle a,\, p\rangle$. Finally $\langle a,\, p\rangle = \langle a,\, x - a\rangle = \langle a,\, x\rangle - \|a\|^2$. □

Two lines. *Write them in a form you can reproduce*: S6 repeats this step with C a cone, M4 repeats it with C a perturbation epigraph, and M11 repeats it with C the constraint cone of a linear program. The point of the module is that they repeat *this* step.

Corollary 15 (The dual description exists)**.** *A nonempty closed convex set is the intersection of all the closed halfspaces containing it.*

Proof. Let D be that intersection. $C \subseteq D$ by definition. If $x \notin C$, Theorem 14 gives a halfspace $\{y : \langle a,\, y\rangle \leq \alpha\}$ containing C but not x, so $x \notin D$. Hence $D \subseteq C$. □

Theorem 16 (Supporting hyperplane)**.** *Let C be convex and $x_0 \in \operatorname{bd}(\operatorname{cl} C)$. Then there is $a \neq 0$ with $\langle a,\, c\rangle \leq \langle a,\, x_0\rangle$ for every $c \in C$.*

Proof. Since x_0 is a boundary point of $\operatorname{cl} C$, there are $x_k \notin \operatorname{cl} C$ with $x_k \to x_0$. Put $a_k = (x_k - P_{\operatorname{cl} C} x_k)/\|x_k - P_{\operatorname{cl} C} x_k\|$, a unit vector; by Theorem 14 applied to $\operatorname{cl} C$, $\langle a_k,\, c\rangle \leq \langle a_k,\, x_k\rangle$ for all $c \in \operatorname{cl} C$. The unit sphere of $\mathbb{R}^n$ is compact — *this is the second and last use of finite dimension in the module* — so a subsequence has $a_k \to a$ with $\|a\| = 1$. Passing to the limit in the inequality gives $\langle a,\, c\rangle \leq \langle a,\, x_0\rangle$ for all $c \in \operatorname{cl} C \supseteq C$. □

Theorem 17 (Strong separation of two sets)**.** *Let C be compact convex, D closed convex, both nonempty, with $C \cap D = \emptyset$. Then there are $a \neq 0$ and $\gamma > 0$ with $\sup_{c\in C}\langle a,\, c\rangle \leq \inf_{d\in D}\langle a,\, d\rangle - \gamma$.*

Proof. Let $E = C - D = \{c - d : c \in C,\ d \in D\}$. It is convex, and it is closed by Proposition 9, whose hypothesis holds because C is bounded so $0^+C = \{0\}$. Since $C \cap D = \emptyset$ we have $0 \notin E$. Apply Theorem 14 to E and the point 0: with $b = 0 - P_E 0 = -P_E 0 \neq 0$,

$$\langle b,\, e\rangle \leq \langle b,\, 0\rangle - \|b\|^2 = -\|b\|^2 \qquad \text{for all } e \in E.$$

Writing $e = c - d$ and setting $a = b$, $\gamma = \|b\|^2 > 0$ gives $\langle a,\, c\rangle \leq \langle a,\, d\rangle - \gamma$ for all $c \in C$, $d \in D$, which is the claim. □

The compactness in Theorem 17 is not decoration. Here is what is lost without it.

Example 9 (Two disjoint closed convex sets that cannot be strongly separated)**.** Take C and D from Example 7: $C = \{(u, v) : u > 0, v \geq 1/u\}$ and $D = \{0\} \times \mathbb{R}$. Both are closed and convex, and they are disjoint since every point of C has $u > 0$. By the computation in Example 7, $C - D$ is the open right half-plane. Suppose $a = (a_1, a_2)$ separates: then $\sup_{e \in C-D} \langle a,\, e\rangle < \infty$. Since the second coordinate of e ranges over all of $\mathbb{R}$, this forces $a_2 = 0$; since the first ranges over $(0, \infty)$, it forces $a_1 \leq 0$. Then $\sup_{e \in C-D} \langle a,\, e\rangle = 0 = \langle a,\, 0\rangle$: the supremum is attained in the limit and the gap is exactly 0. So C and D can be separated *properly* — there is a hyperplane with C weakly on one side and D weakly on the other — but never *strongly*, for any a whatever.

For completeness, the theorem that says exactly when proper separation is available; it is the only statement in this document used without proof besides Proposition 7(2).

Theorem 18 (Proper separation; stated without proof)**.** *Two nonempty convex sets C, D can be properly separated if and only if* $\operatorname{ri} C \cap \operatorname{ri} D = \emptyset$.

This is why Slater's condition in M4 is a statement about relative interiors, and why S2 had to come before S4.

Worked example

Let $C = \bar{B}(0, 1) \subseteq \mathbb{R}^2$ and $x = (3, 0)$. Then $P_C x = (1, 0)$, so $a = x - P_C x = (2, 0)$ and $\|a\|^2 = 4$. Theorem 14 gives $\alpha = \langle a,\, x\rangle - \|a\|^2 = 6 - 4 = 2$, so the separating hyperplane is $\{y : 2y_1 = 2\}$, i.e. $y_1 = 1$: it is the tangent line to the ball at the projection. The predicted gap is $\langle a,\, x\rangle - \alpha = 4 = \mathrm{d}_C(x)^2 = (3 - 1)^2$. *Check this equality in the code, not merely that some separation occurred*: many wrong implementations separate correctly and get the margin wrong, and the margin is the part that carries the information.

Now the same construction for the polytope $C = \operatorname{conv}\{(0, 0), (1, 0), (0, 1)\}$ and $x = (1, 1)$. The nearest point of C is on the edge $u + v = 1$: $P_C x = (0.5, 0.5)$, so $a = (0.5, 0.5)$, $\|a\|^2 = 0.5$, and $\alpha = \langle a,\, x\rangle - 0.5 = 1 - 0.5 = 0.5$. The hyperplane is $0.5(y_1 + y_2) = 0.5$, i.e. $y_1 + y_2 = 1$ — the facet of the triangle that x violates. Note what happened: *the separation construction recovered a facet of the polytope*, which is how the same idea will produce cutting planes in M14.

Where this is used later

Theorem 14 is used in S5 to prove the bipolar theorem and in S6 to prove Farkas' lemma; those two uses are the whole remainder of this module. Beyond it: M4 proves strong duality by separating $(0, p^\star)$ from the perturbation epigraph — literally this theorem; M11 obtains LP duality the same way; M14 generates Benders and Gomory cuts by separating the current point from the convex hull of the feasible set. Example 9 is the mechanism of a nonzero duality gap and of nothing else.

Proof of concept: `separate(C, x)`

Return (a, α) with $a = x - P_C x$ and $\alpha = \langle a,\, x\rangle - \|a\|^2$, built directly on S3's `project`. It is ten lines; the session's work is the mathematics.

Validation.

1. **The margin is exactly the squared distance.** For each set of S3 and 10^3 exterior x: $\langle a,\, x\rangle - \alpha = \|x - P_C x\|^2$ to 10^{-12}. This is the constructive content of Theorem 14, and it is a far stronger check than confirming that some separation occurred.
2. $\sup\langle a,\, c\rangle \le \alpha + 10^{-12}$ over 10^4 sampled $c \in C$, for every set and every x.
3. Reproduce both worked examples exactly: the ball case must return $a = (2, 0)$, $\alpha = 2$, margin 4; the triangle case $a = (0.5, 0.5)$, $\alpha = 0.5$, margin 0.5, and the returned hyperplane must coincide with the violated facet $y_1 + y_2 = 1$ up to positive scaling.
4. Corollary 15, quantitatively: for C a random polygon in $\mathbb{R}^2$, generate the halfspaces produced by m exterior points sampled on a large circle, intersect them, and estimate the area of the intersection by Monte Carlo. Report the areas at $m = 10, 10^2, 10^3, 10^4$ and confirm they decrease monotonically towards the area of C from above.
5. **The compactness hypothesis of Theorem 17, failing.** For C and D of Example 9, maximize the margin $\inf_{d\in D}\langle a,\, d\rangle - \sup_{c\in C}\langle a,\, c\rangle$ over unit a sampled on the circle. Confirm the best margin found is 0 to the sampling resolution and never positive, so no strong separation exists although the sets are disjoint, closed and convex. **Keep this: a nonzero duality gap in M4 is this failure and no other.**

Deliverable. `s4.py` and `separate` in `m1lib.py`.

Exercises

E4.1 Reproduce Theorem 14 with the document closed, marking where closedness and where convexity are used — both are inherited from Theorem 10, and you should be able to say which line of *that* proof each comes from.

E4.2 Prove Corollary 15, and explain with one picture and one sentence why it fails for a nonconvex closed set. This corollary is the reason a convex set has a dual description at all, and every duality in the plan is a dual description of something.

E4.3 Prove Theorem 16, and say precisely why the normalized separating vectors have a convergent subsequence.

E4.4 Verify Example 9 in full and name the failing hypothesis of Theorem 17. **[Reused in M4: this is the duality gap.]**

E4.5 Build the table you will need in S7: three rows — proper, strong, supporting — and three columns — hypothesis, conclusion, counterexample when the hypothesis is dropped. Half a page.

Session gate. Checks 1, 2, 3 and 5 pass; E4.1 and E4.4 are written out.

10 Session S5. Cones, polars and Moreau's decomposition

The question. Separation gives an inequality $\langle a, c\rangle \le \alpha$ with some number α on the right. For a general convex set that number carries information and must be computed. But a cone is scale-invariant: if c is in it, so is $10^6 c$. What does a bounded linear functional on a cone look like, and what does separation become when the number on the right can only be 0?

The picture. If $\langle a, c\rangle > 0$ for a single point c of a cone K, then $\langle a, tc\rangle \to +\infty$, so the supremum of $\langle a, \cdot\rangle$ over K is either 0 or $+\infty$ — never anything in between. The vectors for which it is 0 form a set, the polar cone K°, and separation from a cone always produces a member of it. *This is why cones give exact dualities where general convex sets give only inequalities*, and it is the reason linear programming has no duality gap while general convex programming needs a hypothesis.

The second picture is a decomposition. For a subspace L, every vector splits uniquely as something in L plus something in $L^\perp$, and the two pieces are the two projections. Replace the subspace by a closed convex cone and the orthogonal complement by the polar cone: the same statement is true, with orthogonality of the two pieces surviving intact. That is Moreau's theorem, and it costs three lines given S3.

Statements and proofs

Recall Definition 4: $S^\circ = \{y : \langle y, s\rangle \le 0 \ \forall s \in S\}$. Note that S° is always a closed convex cone, being an intersection of halfspaces through the origin, whatever S is.

Theorem 19 (Bipolar)**.** *For any nonempty $S \subseteq \mathbb{R}^n$, $S^{\circ\circ} = \operatorname{cl}\operatorname{conv}\operatorname{cone}(S)$. In particular a nonempty set K satisfies $K^{\circ\circ} = K$ if and only if K is a closed convex cone.*

Proof. Write $K = \operatorname{cl}\operatorname{conv}\operatorname{cone}(S)$; it is a nonempty closed convex cone. First, $S^\circ = K^\circ$: the inclusion $K^\circ \subseteq S^\circ$ holds since $S \subseteq K$; conversely if $\langle y, s\rangle \le 0$ for all $s \in S$ then $\langle y, \cdot\rangle \le 0$ on all nonnegative combinations of elements of S by linearity, and on their closure by continuity, so $y \in K^\circ$.

($\supseteq$) Let $x \in K$ and $y \in S^\circ = K^\circ$. Then $\langle y, x\rangle \le 0$, and since y was arbitrary in S°, $x \in S^{\circ\circ}$.

($\subseteq$) Let $x \notin K$. Since K is nonempty closed convex, Theorem 14 gives $a \ne 0$ with $\langle a, k\rangle \le \langle a, x\rangle - \|a\|^2$ for all $k \in K$; in particular $\sup_{k\in K}\langle a, k\rangle < \infty$. If $\langle a, k_0\rangle > 0$ for some $k_0 \in K$, then $tk_0 \in K$ for all $t > 0$ and $\langle a, tk_0\rangle \to +\infty$, a contradiction. Hence $\langle a, k\rangle \le 0$ for all $k \in K$, that is, $a \in K^\circ = S^\circ$. Also $0 \in K$, so $0 \le \langle a, x\rangle - \|a\|^2$, giving $\langle a, x\rangle \ge \|a\|^2 > 0$. Thus $a \in S^\circ$ with $\langle a, x\rangle > 0$, so $x \notin S^{\circ\circ}$.

For the last sentence: if $K^{\circ\circ} = K$ then K is a closed convex cone because every polar is; conversely if K is a nonempty closed convex cone then $\operatorname{cl}\operatorname{conv}\operatorname{cone}(K) = K$. □

Theorem 20 (Moreau decomposition)**.** *Let $K \subseteq \mathbb{R}^n$ be a closed convex cone. Every $x \in \mathbb{R}^n$ has a unique decomposition*

$$x = u + v, \qquad u \in K, \quad v \in K^\circ, \quad \langle u, v\rangle = 0,$$

and it is given by $u = P_K x$, $v = P_{K^\circ} x$.

Proof. Construction. Let $p = P_K x$ and $w = x - p$. By Theorem 10(2), $\langle w, y - p\rangle \le 0$ for all $y \in K$. Taking $y = 0 \in K$ gives $\langle w, p\rangle \ge 0$; taking $y = 2p \in K$ (a cone) gives $\langle w, p\rangle \le 0$. Hence $\langle w, p\rangle = 0$. Then for any $y \in K$,

$$\langle w, y\rangle = \langle w, y - p\rangle + \langle w, p\rangle \le 0,$$

so $w \in K^\circ$. Thus $x = p + w$ with $p \in K$, $w \in K^\circ$, $\langle p, w\rangle = 0$.

w *is the projection onto* K°. For any $z \in K^\circ$,

$$\langle x - w,\, z - w\rangle = \langle p,\, z - w\rangle = \langle p,\, z\rangle - \langle p,\, w\rangle = \langle p,\, z\rangle \le 0,$$

the last step because $p \in K$ and $z \in K^\circ$. By Theorem 10(2), $w = P_{K^\circ} x$.

Uniqueness. Suppose $x = u + v$ with $u \in K$, $v \in K^\circ$, $\langle u,\, v\rangle = 0$. For $y \in K$,

$$\langle x - u,\, y - u\rangle = \langle v,\, y\rangle - \langle v,\, u\rangle = \langle v,\, y\rangle \le 0,$$

so $u = P_K x$ by Theorem 10(2), and then $v = x - u$ is determined. □

Corollary 21. *If $K = L$ is a subspace then $L^\circ = L^\perp$ and Theorem 20 is the orthogonal decomposition $x = P_L x + P_{L^\perp} x$.*

Proof. For a subspace, $l \in L$ implies $-l \in L$, so $\langle y,\, l\rangle \le 0$ for all $l \in L$ forces $\langle y,\, l\rangle = 0$; hence $L^\circ = L^\perp$. □

Proposition 22 (Polars of the cones we need)**.** *1.* $(\mathbb{R}^n_+)^\circ = \mathbb{R}^n_-$.
2. $\operatorname{cone}(A)^\circ = \{y : A^\top y \le 0\}$ *for any* $A \in \mathbb{R}^{m\times n}$.
3. The second-order cone $K = \{(z,t) \in \mathbb{R}^{n-1} \times \mathbb{R} : \|z\| \le t\}$ *satisfies* $K^\circ = -K$*; equivalently* $K^* = K$*, so* K *is self-dual.*

Proof. (1) If $y \le 0$ then $\langle y,\, x\rangle \le 0$ for $x \ge 0$. Conversely taking $x = e_i$ gives $y_i \le 0$.

(2) $y \in \operatorname{cone}(A)^\circ$ iff $\langle y,\, Ax\rangle \le 0$ for all $x \ge 0$, iff $\langle A^\top y,\, x\rangle \le 0$ for all $x \ge 0$, iff $A^\top y \le 0$ by part (1).

(3) Let $(w, s) \in K^*$, i.e. $\langle w,\, z\rangle + st \ge 0$ for all $(z,t) \in K$. Taking $t = 1$ and $z = -w/\|w\|$ (legitimate since $\|z\| = 1 \le t$) gives $-\|w\| + s \ge 0$, i.e. $s \ge \|w\|$, so $(w, s) \in K$. Conversely if $\|w\| \le s$ and $\|z\| \le t$ then by Cauchy–Schwarz $\langle w,\, z\rangle + st \ge -\|w\|\|z\| + st \ge -st + st = 0$. Hence $K^* = K$. □

Remark. The companion identity $\{x : Ax \le 0\}^\circ = \operatorname{cone}(A^\top)$ is *not* proved here. By part (2) applied to $A^\top$ we have $\{x : Ax \le 0\} = \operatorname{cone}(A^\top)^\circ$, so taking polars and using Theorem 19 gives $\{x : Ax \le 0\}^\circ = \operatorname{cl}\operatorname{cone}(A^\top)$ — and the closure can be removed only once one knows that $\operatorname{cone}(A^\top)$ is closed. That is Lemma 24, and the resulting identity is Farkas' lemma. *So the missing half of this proposition is exactly the agenda of S6.*

Definition 10 (Normal cone)**.** For C convex and $x \in C$, $N_C(x) = \{v : \langle v,\, y - x\rangle \le 0\ \forall y \in C\}$. By Theorem 10(2), $v \in N_C(x)$ if and only if $x = P_C(x + v)$.

Proposition 23 (The normal cone of the simplex)**.** *For* $x \in \Delta_n$, $N_{\Delta_n}(x) = \{v :\ \{i : x_i > 0\} \subseteq \arg\max_j v_j\}$.

Proof. $v \in N(x)$ iff $\langle v,\, y\rangle \le \langle v,\, x\rangle$ for all $y \in \Delta_n$, iff $\max_{y\in\Delta_n}\langle v,\, y\rangle = \langle v,\, x\rangle$. The maximum of a linear functional over the simplex is $\max_j v_j$, attained exactly at the distributions supported on $\arg\max_j v_j$. So the condition is that x is supported there. □

Worked example

Take $K = \mathbb{R}^2_+$ and $x = (3, -2)$. Then $P_K x = (3, 0)$ by componentwise clipping, and $K^\circ = \mathbb{R}^2_-$ with $P_{K^\circ} x = (0, -2)$. Indeed $(3, 0) + (0, -2) = (3, -2) = x$ and $\langle (3, 0),\, (0, -2)\rangle = 0$: Theorem 20 verified by hand.

Now the normal cone of Δ_3, from Proposition 23. At the interior point $x = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$ the support is everything, so $\arg\max_j v_j$ must be all of $\{1, 2, 3\}$: $N(x) = \{v : v_1 = v_2 = v_3\} = \mathbb{R} \cdot \mathbf{1}$, a line. At the vertex $x = e_1$ the support is $\{1\}$, so the condition is $v_1 \ge v_2$ and $v_1 \ge v_3$: $N(e_1)$ is a *full-dimensional* cone in $\mathbb{R}^3$. At the edge point $x = (\frac{1}{2}, \frac{1}{2}, 0)$ the condition is $v_1 = v_2 \ge v_3$, a two-dimensional cone. The normal cone grows as the face shrinks, which is the general pattern and the reason faces and duality are the same subject.

Where this is used later

Theorem 19 and Proposition 22(2) are the two ingredients S6 combines with Lemma 24 to get Farkas. Definition 10 is how M4 states optimality — $-\nabla f(x^\star) \in N_C(x^\star)$ — and how M8 states stationarity of the projected gradient method. Theorem 20 is the conic case of the general decomposition $x = \operatorname{prox}_f(x) + \operatorname{prox}_{f^*}(x)$ that M3 proves for an arbitrary closed convex function; meeting it here costs three lines and makes M3's version recognizable.

Reinforcement learning. Proposition 23 is why stationarity of a projected policy-gradient method is a set inclusion rather than an equation. At a deterministic policy — a vertex of the simplex — the normal cone is full-dimensional, so an entire cone of gradients is consistent with stationarity: the method can *stop* at a deterministic policy, and generically does. A softmax parameterization has no boundary and therefore no normal cones, and approaches determinism only as its parameters diverge. The same contrast in a different guise — Euclidean versus entropic geometry — is the subject of M9.

Proof of concept: polars and the conic decomposition

Implement `polar` for a polyhedral cone in either representation, using Proposition 22(2), and `moreau_decompose(K, x)` using S3's `project`.

Validation.

1. Theorem 20 for $K = \mathbb{R}^n_+$, the second-order cone, a random polyhedral cone, and a random subspace, over 10^4 random x: $\|x - P_K x - P_{K^\circ} x\| \le 10^{-10}$ and $|\langle P_K x,\, P_{K^\circ} x\rangle| \le 10^{-12}$. For the subspace, additionally $P_K + P_{K^\perp} = I$ to 10^{-14} and both maps are linear, checked as in S3 check 5 — which is Corollary 21.
2. Proposition 22(3): over 10^5 random pairs y, c in the second-order cone, $\langle y,\, c\rangle \ge -10^{-12}$; and the same for $\mathbb{R}^n_+$.
3. Theorem 19, computationally: take a random cone by generators, compute its polar as an inequality description, then the bipolar back as generators, and confirm the two generator sets span the same cone by testing membership in both directions with `linprog` to 10^{-9}. *Include one cone that is not full-dimensional and one that is not pointed*, since those are the cases where a careless implementation silently drops a lineality direction.
4. Reproduce the worked example: N_{Δ_3} at an interior point, a vertex and an edge point, verified by sampling v and checking $\langle v,\, y - x\rangle \le 10^{-12}$ over $y \in \Delta_3$. Report the dimension of each normal cone, estimated from the rank of a sample of its members; the answers must be 1, 3 and 2.

Deliverable. `s5.py`, with `polar` and `moreau_decompose` in `m1lib.py`.

Exercises

E5.1 Reproduce Theorem 19 with the document closed, isolating the sentence in which the cone property — "the supremum is 0 or $+\infty$" — is used.

E5.2 Reproduce Theorem 20 and deduce Corollary 21. **[This is the orthogonal decomposition of MMC201's M2.]**

E5.3 Prove Proposition 22 in full, from Definition 4, with no appeal to a table.

E5.4 Write out the Remark after Proposition 22 as a short argument, ending in the sentence: "Farkas' lemma is Theorem 19 applied to cone(A), and the only missing ingredient is that cone(A) is closed." **[This is the agenda of S6.]**

E5.5 Prove Proposition 23 and compute $N_{\Delta_n}(x)$ at a general point of a proper face. **[Reused in M8.]**

Session gate. Checks 1, 3 and 4 pass; E5.2 and E5.4 are written out.

11 Session S6. Closed cones and Farkas' lemma

The question. You are handed A and b and told that $Ax = b,\ x \geq 0$ has no solution. How would you *prove* it to someone who does not trust you? Exhibiting your failure to find x proves nothing. What is wanted is a short certificate: an object that can be checked in a few arithmetic operations and that makes the nonexistence evident. Does one always exist?

The picture. "$Ax = b$ for some $x \geq 0$" says exactly $b \in \operatorname{cone}(A)$. So the question is geometric: b is outside a cone, and we want a certificate of outsideness. S4 supplies one — a separating hyperplane — and S5 says that for a cone the hyperplane passes through the origin, so the certificate is a single vector y with $\langle y,\, c\rangle \leq 0$ for all $c \in \operatorname{cone}(A)$, i.e. $A^\top y \leq 0$, and $\langle y,\, b\rangle > 0$. Flip signs and that is the second alternative.

But there is a gap in that argument, and it is the whole content of this session. Theorem 14 requires $\operatorname{cone}(A)$ to be *closed*, and by S2 we know that images of closed sets under linear maps are not closed in general — $\operatorname{cone}(A)$ is precisely the image of $\mathbb{R}^n_+$ under A. So the geometry gives the certificate only after one proves that this particular image is closed. That is Lemma 24, it is true because there are only *finitely many* generators, and it is the reason Farkas' lemma is a statement that exactly one of two systems holds rather than merely that they cannot both hold.

Statements and proofs

Lemma 24 (Finitely generated cones are closed)**.** *For any $A \in \mathbb{R}^{m\times n}$, the cone* $\operatorname{cone}(A) = \{Ax : x \geq 0\}$ *is closed.*

Proof. For a subset $I \subseteq \{1, \ldots, n\}$ of column indices, write A_I for the submatrix of those columns, and let $\mathcal{I}$ be the collection of I for which the columns of A_I are linearly independent. By Corollary 3, every element of $\operatorname{cone}(A)$ lies in $\operatorname{cone}(A_I)$ for some $I \in \mathcal{I}$, so

$$\operatorname{cone}(A) = \bigcup_{I\in\mathcal{I}} \operatorname{cone}(A_I).$$

Fix $I \in \mathcal{I}$ and let $A_I x_k \to w$ with $x_k \geq 0$. Since the columns of A_I are independent, A_I is injective, so $A_I^\top A_I$ is invertible and $x_k = (A_I^\top A_I)^{-1} A_I^\top (A_I x_k) \to (A_I^\top A_I)^{-1} A_I^\top w =: x_\infty$ by continuity. Then $x_\infty \geq 0$ as a limit of nonnegative vectors, and $A_I x_\infty = w$ by continuity; so $w \in \operatorname{cone}(A_I)$ and $\operatorname{cone}(A_I)$ is closed. Finally $\mathcal{I}$ is *finite*, and a finite union of closed sets is closed. □

Remark. Two finiteness facts were used: finitely many index subsets, and injectivity of A_I obtained from linear independence. Both fail for a cone generated by infinitely many vectors, and the conclusion fails with them — S7 exhibits a convex cone that is not closed. In M4 the cone in question is a tangent cone, which need not be polyhedral; *a constraint qualification is precisely a hypothesis restoring the conclusion of this lemma*, and that is the single most useful thing to carry from S6 into M4.

Theorem 25 (Farkas)**.** *Let $A \in \mathbb{R}^{m\times n}$ and $b \in \mathbb{R}^m$. Exactly one of the following systems has a solution:*

(I) $x \in \mathbb{R}^n$ *with* $Ax = b$ *and* $x \geq 0$*;*
(II) $y \in \mathbb{R}^m$ *with* $A^\top y \geq 0$ *and* $\langle b,\, y\rangle < 0$*.*

Proof. Not both. Suppose x solves (I) and y solves (II). Then

$$\langle b,\, y\rangle = \langle Ax,\, y\rangle = \langle x,\, A^\top y\rangle \geq 0,$$

since $x \geq 0$ and $A^\top y \geq 0$, contradicting $\langle b,\, y\rangle < 0$. *This is one line, it uses nothing but the definitions, and it is weak duality in its simplest form.*

At least one. Suppose (I) has no solution, i.e. $b \notin \operatorname{cone}(A)$. By Lemma 24 the set $\operatorname{cone}(A)$ is closed; it is convex and nonempty; so Theorem 14 applies and yields $a = b - P_{\operatorname{cone}(A)}b \neq 0$ with

$$\langle a,\, c\rangle \;\leq\; \langle a,\, b\rangle - \|a\|^2 \qquad \text{for every } c \in \operatorname{cone}(A).$$

In particular $\sup_{c\in\operatorname{cone}(A)}\langle a,\, c\rangle < \infty$, and since $\operatorname{cone}(A)$ is a cone this supremum is 0 (as in the proof of Theorem 19: a single c with $\langle a,\, c\rangle > 0$ would make it $+\infty$). Taking $c = 0$ in the display gives $0 \leq \langle a,\, b\rangle - \|a\|^2$, so $\langle a,\, b\rangle \geq \|a\|^2 > 0$. And $\langle a,\, Ax\rangle \leq 0$ for all $x \geq 0$ means $A^\top a \leq 0$. Now put $y = -a$: then $A^\top y \geq 0$ and $\langle b,\, y\rangle = -\langle a,\, b\rangle \leq -\|a\|^2 < 0$, so y solves (II). □

The one line that matters. In the second half of that proof, everything is a definition chase except the appeal to Theorem 14, and that appeal is legitimate *only because of Lemma 24.* Mark it. The entire difference between the two halves of the proof is that the first needs nothing and the second needs the set to be closed — which is the same as saying that weak duality is free and strong duality is not.

Corollary 26 (Inequality form)**.** *Exactly one of the following has a solution: x with $Ax \leq b$; or $y \geq 0$ with $A^\top y = 0$ and $\langle b,\, y\rangle < 0$.*

Proof. Write $x = x^+ - x^-$ with $x^\pm \geq 0$ and introduce a slack $s \geq 0$: the system $Ax \leq b$ is solvable iff

$$\begin{bmatrix} A & -A & I \end{bmatrix} \begin{bmatrix} x^+ \\ x^- \\ s \end{bmatrix} = b, \qquad (x^+, x^-, s) \geq 0$$

is solvable. Apply Theorem 25 to $\tilde{A} = [A \;\; -A \;\; I]$: the alternative is y with $\tilde{A}^\top y \geq 0$ and $\langle b,\, y\rangle < 0$, and $\tilde{A}^\top y \geq 0$ reads $A^\top y \geq 0$, $-A^\top y \geq 0$, $y \geq 0$, i.e. $A^\top y = 0$ and $y \geq 0$. □

Theorem 27 (Gordan)**.** *Exactly one of the following has a solution: x with $Ax < 0$ (strictly, componentwise); or $y \geq 0$, $y \neq 0$, with $A^\top y = 0$.*

Proof. First, $Ax < 0$ has a solution iff $Ax \leq -\mathbf{1}$ has one: if $Ax < 0$, put $\sigma = \min_i(-(Ax)_i) > 0$ and use x/σ; conversely $Ax \leq -\mathbf{1}$ gives $Ax < 0$. Apply Corollary 26 with right-hand side $-\mathbf{1}$: exactly one of $Ax \leq -\mathbf{1}$, and $y \geq 0$ with $A^\top y = 0$, $\langle -\mathbf{1},\, y\rangle < 0$. The last condition is $\sum_i y_i > 0$, which for $y \geq 0$ is the same as $y \neq 0$. □

Stiemke's and Motzkin's theorems have the same shape and are obtained the same way; the pattern worth carrying forward is that *each theorem of the alternative is Theorem 25 for a different cone, and the cone can be read off the statement.* That is worth more than memorizing four statements.

Worked example

Let

$$A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}, \qquad b = \begin{bmatrix} 1 \\ 2 \end{bmatrix}.$$

Here $\operatorname{cone}(A) = \{s(1,1) : s \geq 0\}$, a ray, and b is not on it, so (I) has no solution. Run the proof and extract the certificate.

The projection of b onto the ray in direction $\hat{u} = (1,1)/\sqrt{2}$ is $Pb = \langle b,\, \hat{u}\rangle\hat{u} = \frac{3}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}}(1,1) = (1.5, 1.5)$ — the inner product $\frac{3}{\sqrt{2}}$ is positive, so the projection lands in the ray rather than at the apex. Hence

$$a = b - Pb = (-0.5,\; 0.5), \qquad y = -a = (0.5,\; -0.5).$$

Check the certificate: $A^\top y = (0.5 - 0.5,\ 0.5 - 0.5) = (0, 0) \geq 0$ ✓, and $\langle b,\, y\rangle = 0.5 - 1 = -0.5 < 0$ ✓. So y proves that no $x \geq 0$ has $Ax = b$.

Check the margin: $-\langle b,\, y\rangle = 0.5$ and $\|b - Pb\|^2 = 0.25 + 0.25 = 0.5$. *They agree, and they must*, because this is Theorem 14's margin identity carried through Theorem 25. Verifying that equality in code is the single most informative check in this module: it tests the proof rather than the statement.

Where this is used later

Theorem 25 is quoted three more times in the plan, each time with $\text{cone}(A)$ replaced by another closed convex cone:

- in **M4**, with the linearized cone of the active constraints, giving the Karush–Kuhn–Tucker conditions — and, as the Remark after Lemma 24 says, a constraint qualification is exactly the hypothesis that lets the argument run;
- in **M11**, with the constraint cone of a linear program, giving strong duality with no qualification needed, because Lemma 24 supplies closedness for free in the polyhedral case. *That sentence is the reason linear programming duality is cleaner than convex duality*;
- in **M14**, with the dual feasible cone of a Benders subproblem, where the extreme rays produced by the alternative are the feasibility cuts.

Reinforcement learning. In M11 the Bellman inequalities $V \geq r_a + \gamma P_a V$ define the feasible set of the linear programming formulation of a Markov decision process. Theorem 25 is what certifies that a modified system — for instance one with an added constraint encoding a safety budget — is infeasible, and the certificate y is, up to normalization, an occupancy measure. Infeasibility certificates and occupancy measures are the same objects seen from two sides.

Proof of concept: the Farkas certificate

Implement `farkas_certificate(A, b)` returning either $x \geq 0$ with $Ax = b$, or y with $A^\top y \geq 0$ and $\langle b,\, y\rangle < 0$ — never both and never neither.

Validation.

1. On 10^3 random feasible instances (draw $x_0 \geq 0$, set $b = Ax_0$): the routine returns $x \geq -10^{-12}$ with $\|Ax - b\| \leq 10^{-9}$, and reports no certificate.
2. On 10^3 random infeasible instances (draw y_0, choose A with $A^\top y_0 \geq 0$ and b with $\langle b,\, y_0\rangle < 0$): the returned y satisfies $A^\top y \geq -10^{-10}$ componentwise and $\langle b,\, y\rangle \leq -10^{-6}$, and independently `scipy.optimize.linprog` reports the primal infeasible. *Both directions must be checked on the same instance;* a routine that only ever confirms the branch it took has not been tested.
3. **The certificate is the projection.** Verify that $y = -(b - P_{\text{cone}(A)}b)$, computed with the S3 and S5 code, is a valid certificate, and that its margin $-\langle b,\, y\rangle$ equals $\|b - P_{\text{cone}(A)}b\|^2$ to 10^{-9}. *This is the proof of Theorem 25 executed rather than read.*
4. Reproduce the worked example exactly: $Pb = (1.5, 1.5)$, $y = (0.5, -0.5)$, $A^\top y = (0, 0)$, $\langle b,\, y\rangle = -0.5$, margin 0.5.
5. Theorem 27: over 10^3 random A, exactly one of the two systems is solvable, with zero disagreements between the two independent solves.

Deliverable. `s6.py` and `farkas_certificate` in `m1lib.py`. **Keep it:** M11 asks for the same routine as the infeasibility certificate of a linear program, and M12's Phase I produces one as a by-product of the simplex method — at which point you should check that the two certificates agree up to positive scaling.

Exercises

E6.1 Reproduce Lemma 24 with the document closed. Which two finiteness facts are used? **[Reused in S7 and, in spirit, in M4.]**

E6.2 Reproduce Theorem 25, writing the "not both" argument separately and in one line.

E6.3 Prove Corollary 26 and Theorem 27, and for each identify the cone to which Theorem 25 was applied.

E6.4 Use Theorem 25 to complete the Remark after Proposition 22: show $\{x : Ax \leq 0\}^\circ = \text{cone}(A^\top)$, with no closure.

E6.5 **The central exercise of the module.** Write the proof of Theorem 25 on one page so that the appeal to Theorem 14 is a single displayed step and the appeal to Lemma 24 is a single marked line. **Keep this page. M4 repeats it with the tangent cone and obtains KKT; M11 repeats it with the constraint cone of a linear program and obtains LP duality. You will be asked in M11 to lay the three proofs side by side, and they must look like one proof.**

Session gate. Checks 2, 3, 4 and 5 pass; E6.1, E6.2 and E6.5 are written out.

12 Session S7. Consolidation, capstone and checkpoint

The question. Three hypotheses have been carried through six sessions: convexity, closedness, and finite generation. Each was used, and each was claimed to be load-bearing. Is that true? What exactly goes wrong when each is dropped — and, since real computation is inexact, what goes wrong when a hypothesis holds but only barely?

The picture. Two failures, one qualitative and one quantitative.

Qualitatively: there are convex cones that are not closed, and on such a cone the bipolar theorem fails, strong separation fails, and Farkas' lemma would be false. Lemma 24 says no *finitely generated* cone can behave this way, which is why Theorem 25 is an alternative. So the qualitative failure is not reachable from a matrix A — it needs infinitely many generators, which is what semi-infinite and general convex programming have, and it is where duality gaps live.

Quantitatively: what *is* reachable from a matrix is near-failure. Make the generators of $\operatorname{cone}(A)$ nearly dependent and put b just outside. Farkas' lemma remains exactly true and the certificate remains exactly valid — but its margin $\mathrm{d}_{\operatorname{cone}(A)}(b)^2$ shrinks towards zero, and below the solver's tolerance the two alternatives become indistinguishable numerically. The theorem is intact and the decision is impossible. *This is the same phenomenon M4 calls a constraint qualification that holds only barely and M13 calls a badly centred starting point*, and meeting it once, here, at a moment when nothing depends on it, is the purpose of the capstone.

The capstone: a convex cone that is not closed

Let

$$K = \{(u, v) \in \mathbb{R}^2 : u > 0\} \cup \{(0, 0)\}.$$

K is a cone: $\lambda(u, v)$ has positive first coordinate for $\lambda > 0$ and $u > 0$, and $\lambda = 0$ gives the origin. K is convex: a convex combination of two points with $u > 0$ has $u > 0$, and combining either with the origin keeps $u \geq 0$ with $u > 0$ unless all the weight is at the origin. K is not closed: $(1/k, 0) \in K$ and $(0, 0) \in K$, but $(0, 1) \notin K$ while $(1/k, 1) \in K$ converges to it.

Compute the polar. If $(p, q) \in K^\circ$ then $pu + qv \leq 0$ for every $u > 0$ and every $v \in \mathbb{R}$; letting $v \to \pm\infty$ forces $q = 0$, and letting $u \to 0^+$ then $u \to \infty$ forces $p \leq 0$. Conversely $(p, 0)$ with $p \leq 0$ is in K°. So

$$K^\circ = \{(p, 0) : p \leq 0\}, \qquad \text{and hence} \qquad K^{\circ\circ} = \{(u, v) : pu \leq 0 \,\forall p \leq 0\} = \{(u, v) : u \geq 0\} = \operatorname{cl} K.$$

So $K^{\circ\circ} = \operatorname{cl} K \supsetneq K$: Theorem 19 holds, in the form it was proved — $S^{\circ\circ} = \operatorname{cl}\operatorname{conv}\operatorname{cone}(S)$ — and the identity $K^{\circ\circ} = K$ fails exactly because K is not closed. Moreover the point $b = (0, 1) \in \operatorname{cl} K \setminus K$ cannot be strongly separated from K: every $a \in K^\circ$ has $\langle a, b\rangle = 0$, so no hyperplane puts b strictly on one side. *Farkas' lemma would be false for this cone*, and by Lemma 24 no matrix can produce it.

The hypothesis table

Reproduce and complete this table from memory as part of the checkpoint.

Hypothesis	What it buys	What breaks without it
Convexity of C	Uniqueness in Thm. 10(1); the characterization in Thm. 10(2)	The nearest point need not be unique; d_C is no longer convex; Cor. 15 fails, so there is no dual description
Closedness of C	Existence in Thm. 10(1); strong separation, Thm. 14	The infimum is approached only at infinity (Ex. 7); $K^{\circ\circ} \neq K$ (S7 capstone); duality gaps in M4
Compactness of one set	Closedness of $C - D$, hence Thm. 17	Disjoint closed convex sets with no strong separation (Ex. 9)
Finite generation	Lemma 24, hence Thm. 25 as an *alternative*	Non-closed cones exist; the certificate may not; this is what a constraint qualification restores in M4
Finite dimension	Thm. 2; the compactness of the unit sphere in Thm. 16	Carathéodory is false as stated; supporting hyperplanes need a different argument

Proof of concept: the suite, the non-closed cone, and near-degeneracy

Assemble `s1.py` to `s6.py` into one runner that imports `m1lib.py`, executes every validation check of the module with the single module seed, and prints a table of check, value, tolerance and verdict.

Validation.

1. Every check from S1 to S6 passes when run from the single runner with the module seed. *If a check passes only in its own file, something was hard-coded; find it.*
2. For the capstone cone K and $b = (0, 1)$: sample unit vectors a, keep those with $\sup_{c \in K} \langle a,\, c\rangle \leq 0$ (numerically, over a large sample of $c \in K$), and report $\max\langle a,\, b\rangle$ over them. Confirm it is 0 to the sampling resolution, so no strong separation exists; and confirm $b \in K^{\circ\circ}$ using S5's code. **Report both numbers together: this is the closedness hypothesis of Theorems 14 and 19 failing.**
3. Build the near-degenerate family $A_\varepsilon = \begin{bmatrix} 1 & 1 \\ 0 & \varepsilon \end{bmatrix}$ and $b_\varepsilon = (1, -\varepsilon)$. The two generators $(1, 0)$ and $(1, \varepsilon)$ span the thin wedge $\mathrm{cone}(A_\varepsilon) = \{(u, v) : v \geq 0,\ u \geq v/\varepsilon\}$, and b_ε sits just below it, so system (I) is infeasible for *every* $\varepsilon > 0$. Here $P_{\mathrm{cone}(A_\varepsilon)} b_\varepsilon = (1, 0)$ and the Farkas margin is $\|b_\varepsilon - P b_\varepsilon\|^2 = \varepsilon^2$: the certificate $y = (0, \varepsilon)$ is exactly valid at every ε, and its margin decays *quadratically*. For $\varepsilon = 10^{-1}, \ldots, 10^{-6}$, tabulate the computed margin against ε^2 and confirm they agree to solver tolerance until the margin falls below it. Report the smallest ε at which your feasibility solve and your certificate solve disagree about the instance — with a default `linprog` tolerance this happens well before $\varepsilon = 10^{-6}$. **Keep this table: the theorem is exactly true at every ε and the numerical decision becomes impossible anyway. M13 asks why an interior point method needs a well-centred start, and this is the answer.**

Deliverable. `run_m1.py`, its printed verdict table with one row per check for the whole module, and the two tables of checks 2 and 3.

Checkpoint

The module is complete when the following can be produced from scratch, on paper, with this document closed:

1. Theorem 10, all three parts, with proof, and with the line using closedness and the line using convexity marked.
2. Theorem 14, with proof, the separating vector written explicitly as $x - P_C x$, and the margin identified as $\mathrm{d}_C(x)^2$.

3. Lemma 24 and Theorem 25, with proofs, the single line where closedness is used marked, and the one-line "not both" argument written separately.
4. Theorem 20, with proof, and the sentence identifying the orthogonal decomposition of a subspace as its special case.
5. The hypothesis table above, reproduced and with at least one entry of your own added.

Keep item 3. M4 asks the identical question with the tangent cone of the active constraints in place of cone(A), and the answer is the Karush–Kuhn–Tucker theorem; M11 asks it with the constraint cone of a linear program, and the answer is LP duality. The method is Theorem 25 exactly as used here; only the cone changes. Without this proof, KKT will look like a list of conditions someone wrote down, and LP duality like a separate theory.

Keep item 1. M8 uses the fixed-point equation $x = P_C(x - t\nabla f(x))$ as its stationarity condition, and it is Theorem 10(2) restated; M9 uses Theorem 10(3), the firm version, to prove Fejér monotonicity of proximal iterates. Both will look like new material if the projection theorem is not in hand.

Session gate. The runner is green and the checkpoint has been written out on paper. M1 is finished; M2 may be started.

13 Common errors

- **Reading "closed" as boilerplate.** It is the hypothesis that fails, and S7 is built around the failure. Strong separation, the bipolar theorem and Farkas' lemma are all false without it, and every duality gap in the plan is this failure in disguise.
- **Confusing the interior with the relative interior.** Δ_n has empty interior in $\mathbb{R}^n$ (Example 6). Any argument that produces an interior point of a set defined partly by equalities has a bug; the correct statement uses ri.
- **Proving the projection formula by solving the optimization problem.** Once Theorem 10(2) is available, every closed form in Proposition 12 and 13 is verified by *checking an inequality on a guess*. If a derivation in your notes optimizes, it is longer than it needs to be and will not generalize.
- **Expecting P_C to be linear.** It is nonexpansive, not linear, unless C is affine. The consequence in M8 is that a projected gradient step is not a gradient step followed by a separable correction. S3 check 5 exists to make this a number.
- **Clipping and renormalizing instead of projecting onto the simplex.** The worked example of S3 gives a discrepancy of 0.248 on a four-dimensional instance. This is the single most common implementation error in projected policy-gradient code.
- **Saying "separating hyperplane" without saying which separation.** Proper, strong and supporting are three theorems with three hypotheses. Which one is available determines whether the duality gap of M4 is zero. E4.5 exists for this.
- **Checking that a separation occurred rather than checking the margin.** Theorem 14 predicts the gap exactly: $\mathrm{d}_C(x)^2$. An implementation that separates with the wrong margin has a bug that the weaker check will never find. S4 check 1 and S6 check 3 are the two places this matters.
- **Mixing the polar and the dual cone.** $C^* = -C^\circ$. Both conventions are in print; this document uses the polar throughout. A sign error here propagates silently into every KKT condition in M4. Assert the convention in a docstring in `m1lib.py`.
- **Believing both Farkas alternatives could fail.** They cannot, and the reason is Lemma 24 and nothing else. If an implementation ever reports neither, the tolerance in the certificate check is looser than the tolerance in the feasibility solve — a numerical artifact of exactly the kind S7 check 3 is about, not a counterexample.
- **Treating S1 as the soft session.** Corollary 3 is the proof of Lemma 24, which is the hypothesis of Theorem 25, which is the theorem the rest of the plan quotes. The module's logical chain runs through its shortest session.

14 What M1 hands forward

15 References

Nothing in this list is required reading: every proof above is written out above. Each entry says what it adds.

[1] Hiriart-Urruty, J.-B. and Lemaréchal, C. (2001). *Fundamentals of Convex Analysis*. Springer. Part A. The closest published treatment to this module's sequence, with more examples and the infinite-dimensional caveats.

[2] Boyd, S. and Vandenberghe, L. (2004). *Convex Optimization*. Cambridge University Press. Ch. 2 and §8.1: a large catalogue of convex sets and closed-form projections. Freely available.

[3] Crouzeix, J.-P. and Ocaña, E. (2018). *Análisis convexo*. UNI. The MMC007 syllabus text; use it to fix the Spanish vocabulary of the lectures.

[4] Rockafellar, R. T. (1970). *Convex Analysis*. Princeton University Press. §6, 11, 14, 17, 19, 22. The reference statements later modules cite, and the source of the two proofs omitted here (§6

Table 3: Results established in M1, the session that establishes each, and where each is used again.

Result	**From**	**Used in**	**As**
Projection theorem (Thm. 10)	S3	M8	Stationarity as $x = P_C(x - t\nabla f(x))$, and the gradient mapping
Projection theorem, part 3	S3	M9	Firm nonexpansiveness, hence Fejér monotonicity of proximal iterates
Projection theorem, affine case	S3	MMC201 M2	The normal equations; least squares as an orthogonal projection
$\frac{1}{2}\,\mathrm{d}_C^2$ and P_C (Prop. 11)	S3	M3	The Moreau envelope and proximal operator of the indicator of C
Simplex projection (Prop. 13)	S3	M8, M9, M15	The projected policy-gradient step, at $O(n \log n)$ per state
Strong separation (Thm. 14)	S4	M4	Strong duality under Slater: separate $(0, p^\star)$ from the perturbation epigraph
Failure of strong separation (Ex. 9)	S4	M4	What a nonzero duality gap is
Supporting hyperplane (Thm. 16)	S4	M11, M14	Faces and extreme points of a polyhedron; cutting planes
Proper separation (Thm. 18)	S4	M4	Why Slater's condition is stated with ri
Bipolar theorem (Thm. 19)	S5	M4, M11	Dual descriptions of cones; the dual of a polyhedral constraint set
Moreau decomposition (Thm. 20)	S5	M3	The conic case of $x = \mathrm{prox}_f(x) + \mathrm{prox}_{f^*}(x)$
Normal cone (Def. 10, Prop. 23)	S5	M4, M8	$-\nabla f(x^\star) \in N_C(x^\star)$ as constrained stationarity
Closedness of cone(A) (Lem. 24)	S6	M11	Why linear programming needs no constraint qualification
Farkas' lemma (Thm. 25)	S6	M4	KKT, with the tangent cone in place of cone(A)
Farkas' lemma	S6	M11, M12	LP duality; the Phase I infeasibility certificate
Farkas' lemma	S6	M14	Benders feasibility cuts as extreme rays of the dual subproblem
Fourier–Motzkin (E2.4)	S2	M14	Projecting out the complicating variables in Benders decomposition
Carathéodory (Thm. 2)	S1	M8, M11	Sparsity of Frank–Wolfe iterates; mixtures of deterministic policies

for Prop. 7(2), Thm. 11.3 for Thm. 18).

[5] Bertsekas, D. P. (1999). *Nonlinear Programming*, 2nd ed. Athena Scientific. App. B: recession cones and closedness criteria in far more detail than S2 needs.

[6] Bauschke, H. and Combettes, P. (2017). *Convex Analysis and Monotone Operator Theory in Hilbert Spaces*, 2nd ed. Springer. The operator reading of P_C as a firmly nonexpansive map, which is the language M9 will use.

[7] Schrijver, A. (1986). *Theory of Linear and Integer Programming*. Wiley. §7.3: Farkas' variants collected, proved polyhedrally rather than by separation — worth seeing once as a contrast with S6.

[8] Bertsimas, D. and Tsitsiklis, J. (1997). *Introduction to Linear Optimization*. Athena Scientific. §4.6: Farkas read as an infeasibility certificate, the form M11 and M12 use.

[9] Beck, A. (2017). *First-Order Methods in Optimization*. SIAM. Ch. 6: a larger projection table, and the projection as a special proximal operator, which is M3.

[10] Farkas, J. (1902). Theorie der einfachen Ungleichungen. *J. Reine Angew. Math.* 124, 1–27. Theorem 25 in the original.

[11] Gordan, P. (1873). Über die Auflösung linearer Gleichungen mit reellen Coefficienten. *Math. Ann.* 6, 23–28. Theorem 27 in the original.

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[13] Moreau, J.-J. (1962). Décomposition orthogonale d'un espace hilbertien selon deux cônes mutuellement polaires. *C. R. Acad. Sci. Paris* 255, 238–240. Theorem 20 in the original; two pages.

[14] Duchi, J., Shalev-Shwartz, S., Singer, Y. and Chandra, T. (2008). Efficient projections onto the ℓ_1-ball for learning in high dimensions. *ICML*; and Condat, L. (2016). Fast projection onto the simplex and the ℓ_1 ball. *Mathematical Programming* 158, 575–585. The $O(n \log n)$ algorithm of Proposition 13 and its refinements.

[15] Zhao, S. (2025). *Mathematical Foundations of Reinforcement Learning*. Springer. The expository method this document follows — question, picture, statement, proof — and the natural destination of the reinforcement-learning remarks in each session.